

MATRICES

Points to remember

- The square array of scalars is the matrix, whereas the determinant is just one scalar, associated with the matrix according to a definite rule.
- The trace of a square matrix is given by $\text{tr}(A) = a_{11} + a_{22} + a_{33} + \dots + a_{nn}$
- In the process of multiplication AB we take the rows of A and the columns of B , which is possible when the number of columns of A is same as numbers of rows of B .
 - Matrix multiplication is not commutative, in general.
 - From $AB = AC$ or $BA = CA$ we cannot in general conclude that $B = C$, even if $A \neq O$.
- Let $A = [a_{ij}]_{m \times p}$ and $B = [b_{ij}]_{p \times n}$, then $(AB)' = BA'A'$
- A square matrix A is said to be involutory if $A^2 = I$
 - A square matrix A is called periodic, if $A^{k+1} = A$, where k is a positive integer. If k is the least positive integer for which $A^{k+1} = A$, then k is said to be period of A .
 - If $A^2 = A$, then A is called idempotent matrix.
 - A square matrix A is called nilpotent matrix of order m provided it satisfies the relation $A^k = O$ and $A^{k-1} \neq O$.
- $$|A| = a_{11}M_{11} - a_{12}M_{12} + a_{13}M_{13}$$

$$= a_{11}A_{11} + a_{12}A_{12} + a_{13}A_{13}$$
- The value of a determinant remains unaltered, if the rows and columns are inter changed. i.e. if A is an n -rowed symmetric determinant of odd order is zero.
- If any two rows (or columns) of a determinant be interchanged, the value of determinant is changed in sign only.
- If a determinant has any two rows (or columns) identical or proportional, then its value is zero.
- If all the elements of any row (for columns) be multiplied by the same number, then the determinant is multiplied by that number.
- If each element of any row (or column) can be expressed as a sum of two terms then the determinant can be expressed as the sum of two determinants.
- The value of a determinant is not altered by adding to the elements of any row (or columns) the same multiples of the corresponding elements of any other row (or column).
- If the elements of a determinant Δ are rational integral functions of x and two rows (or columns) become identical when $x = a$, then $x - a$ is a factor of Δ . Again, if r rows becomes identical when a is substituted for x , then $(x - a)^{r-1}$ is a factor of Δ .

$$\begin{vmatrix} 1 & 1 & 1 \\ \alpha & \beta & \gamma \\ \alpha^2 & \beta^2 & \gamma^2 \end{vmatrix} = (\beta - \gamma)(\gamma - \alpha)(\alpha - \beta).$$

$$\begin{vmatrix} 1 & 1 & 1 \\ \alpha & \beta & \gamma \\ \alpha^3 & \beta^3 & \gamma^3 \end{vmatrix} = (\beta - \gamma)(\gamma - \alpha)(\alpha - \beta)(\alpha + \beta + \gamma).$$
- If A and B be two square matrices of the same order, then $|AB| = |A| |B| = |BA|$.
- If $\Delta = |a_{ij}|$, is a determinant of n th order and $\Delta' = |A_{ij}|$, the adjoint of Δ , then $\Delta' = \Delta^{n-1}$ when $\Delta \neq 0$
- If the same number x is added to all elements of the determinant D , the determinant is increased by the product of the number into the sum of the cofactors of all elements of D .

$$D' = D + x \sum_{i=1}^n A_{ij}$$

18. $d\Delta/dx$ is the sum of the n determinants obtained by differentiating the elements of one row (or column) of Δ and leaving the elements of the other $n - 1$ rows (or columns) unaltered.
19. If $A = [a_{ij}]_n$ then cofactor $A = [A_{ij}]_n$ when A_{ij} is the cofactor of $a_{ij} \forall i \& j$.
 $\text{Adj } A = [b_{ij}]_n$ where $b_{ij} = A_{ji} \& i \& j$.
20. If A is a square matrix of order n , then $A \cdot (\text{adj } A) = |A| I_n = (\text{adj } A) \cdot A$.
21. (i) $|\text{adj } A| = |A|^{n-1}$, where n is order of A .
(ii) If $|A| = 0$, then $|\text{adj } A| = 0$
(iii) $\text{adj } (kA) = k^{n-1} (\text{adj } A)$
(iv) $\text{adj } A' = (\text{adj } A)'$
(v) $\text{adj } I = I$
(vi) $\text{adj}(AB) = (\text{adj } B) (\text{adj } A)$.
(vii) If A is a non-singular matrix, then $\text{adj } (\text{adj } A) = |A|^{n-2} A$.
(viii) $|\text{adj } (\text{adj } A)| = |A|^{(n-1)^2}$
22. $A^{-1} \frac{1}{|A|} \text{adj } A$.
23. (i) $(A^{-1})^{-1} = A$, if A is non-singular.
(ii) $|A^{-1}| = \frac{1}{|A|}$
(iii) $(A')^{-1} = (A^{-1})'$
(iv) If $k \neq 0$ and A is a nonsingular matrix then $(kA)^{-1} = \frac{1}{k} A^{-1}$.
(v) $I^{-1} = I$
(vi) If $D = \text{diag}(a_{11}, a_{22}, \dots, a_{nn})$ where $a_{ii} \neq 0 \forall i$ then $D^{-1} = \text{diag}\left(\frac{1}{a_{11}}, \frac{1}{a_{22}}, \dots, \frac{1}{a_{nn}}\right)$.
(vii) We define $A^{-n} = (A^{-1})^n$ when A is non-singular and n is a positive integer. We define $A^0 = I$ for an arbitrary square matrix A , singular or not. Also the laws of exponents $A^m A^n = A^{m+n}$ and $(A^m)^n = A^{mn}$ apply for all integral values of m and n when A is non-singular.
(viii) $\text{adj } A^{-1} = (\text{adj } A)^{-1} = \frac{A}{|A|}$.
(ix) If A, B be two n -rowed non-singular matrices, then AB is also non-singular and $(AB)^{-1} = B^{-1} A^{-1}$.
(x) If A, B are square matrices and $AB = O$, then $A = O$, or $B = O$, or A and B are both singular.
(xi) Let A be a non-singular matrix. Then $AB = AC \Rightarrow B = C$ and $BA = CA \Rightarrow B = C$.
24. The Cayley-Hamilton Theorem : Every square matrix satisfies its characteristic equation
i.e., if for a square matrix A of order n ,
 $|A - \lambda I| = (-1)^n [\lambda^n + a_1 \lambda^{n-1} + a_2 \lambda^{n-2} + \dots + a_n]$, then the matrix equation
 $X^n + a_1 X^{n-1} + a_2 X^{n-2} + \dots + a_n I = O$
is satisfied by $X = A$
i.e., $A^n + a_1 A^{n-1} + a_2 A^{n-2} + \dots + a_n I = O$.
25. A square matrix A is said to be orthogonal if $A'A = I$.
(i) Every orthogonal matrix is nonsingular.
(ii) I is an orthogonal matrix.
(iii) $|A|$ is either 1 or -1 when A is orthogonal.
(iv) The product of two orthogonal matrices is orthogonal.
(v) The inverse of an orthogonal matrix is orthogonal.
(vi) The transpose of an orthogonal matrix is orthogonal.

26. Let $A = [a_{ij}]_{m \times n}$ be a given matrix. A natural number ρ is said to be the rank of A if A has a non singular submatrix of order ρ and it has no non-singular submatrix of order more than ρ . Rank of zero matrix is regarded to be zero.
27. Cramer's rule : If $\Delta \neq 0$, we have

$$x_i = \frac{\Delta_i}{\Delta}, \quad i = 1, 2, \dots, n.$$
where Δ_i is the determinant obtained by replacing the i th column in Δ by the elements $b_1, b_2, \dots, b_n$.
28. **Non-Homogeneous System**
Let $\rho(A)$ denote rank of A and $\rho([AB])$ denote rank of the augmented matrix. Clearly $\rho(A) \leq \rho([AB])$. In all there are three possibilities :
(a) If $\rho(A) < \rho([AB])$ then the system has no solution (i.e. system is inconsistent).
(b) If $\rho(A) = \rho([AB]) = \text{number of unknowns}$, then the system has unique solution. (and hence is consistent)
(c) If $\rho(A) = \rho([AB]) < \text{number of unknowns}$, then the system has infinitely many solutions (and so is consistent).
29. **Homogeneous System**
Let the homogeneous system be $AX = 0$, m equations in ' n ' unknowns. in this case $B = 0$ and so $\rho(A) = \rho([AB])$. Hence if $\rho(A) = n$, then the system has only the trivial solution. If $\rho(A) < n$, then the system has infinitely many solutions.

DETERMINANT

Points to remember

- The symbol $\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}$ is called the determinant of order two.
Its value is given by : $D = a_1 b_2 - a_2 b_1$
- The symbol $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ is called the determinant of order three.
Its value can be found as :

$$D = a_1 \begin{vmatrix} b_2 & c_2 \\ b_3 & c_3 \end{vmatrix} - a_2 \begin{vmatrix} b_1 & c_1 \\ b_3 & c_3 \end{vmatrix} + a_3 \begin{vmatrix} b_1 & c_1 \\ b_2 & c_2 \end{vmatrix} \text{ or}$$

$$D = a_1 \begin{vmatrix} b_2 & c_2 \\ b_3 & c_3 \end{vmatrix} - b_1 \begin{vmatrix} a_2 & c_2 \\ a_3 & c_3 \end{vmatrix} + c_1 \begin{vmatrix} a_2 & b_2 \\ a_3 & b_3 \end{vmatrix} \dots \text{ and so on.}$$
In this manner we can expand a determinant in 6 ways using elements of;
 R_1, R_2, R_3 or C_1, C_2, C_3 .
- following examples short hand writing large expressions are :
(i) The lines : $a_1 x + b_1 y + c_1 = 0 \dots (1)$
 $a_2 x + b_2 y + c_2 = 0 \dots (2)$
 $a_3 x + b_3 y + c_3 = 0 \dots (3)$
are concurrent if, $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0$.
Condition for the consistency of three simultaneous linear equations in 2 variables.
(ii) $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents a pair of straight lines if :

$$abc + 2fgh - af^2 - bg^2 - ch^2 = 0 = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix}$$
(iii) Area of a triangle whose vertices are $(X_r, Y_r); r = 1, 2, 3$, is

$$D = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} \text{ if } D = 0 \text{ then the three points are collinear.}$$

(iv) Equations of a straight line passing through (x_1, y_1) & (x_2, y_2) is $\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0$

4. Minors :

The minors of a given element of a determinant is the determinant of the elements which remain after deleting the row & column in which the given element stands. For example, the minor of a_1 in (Key Concept 2) is $\begin{vmatrix} b_2 & c_2 \\ b_3 & c_3 \end{vmatrix}$

& the minor of b_2 is $\begin{vmatrix} a_1 & c_1 \\ a_3 & c_3 \end{vmatrix}$

Henc a determinant of order two will have “4 minors” & a determinant of order three will have “9 minors”.

5. Cofactor:

If M_1 represents the minor of some typical element then the cofactor is defined as :

$C_{ij} = (-1)^{i+j} \cdot M_{ij}$; Where i & j denoted the row & column in which the particular element lies. Note that the value of a determinant of order three in terms of ‘Minor’ & ‘Cofactor’ can be written as : $D = a_{11}M_{11} - a_{12}M_{12} + a_{13}M_{13}$ or $D = a_{11}C_{11} - a_{12}C_{12} + a_{13}C_{13}$ & so on.....!

6. Properties of Determinants:

P-1: The value of a determinant remains unaltered, if the rows & colms are inter changed. e.g.

$$\text{If } D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = D'$$

D & D' are transpose of each other. If $D' = -D$ then it is Skew symmetric determinant but $D' = D \Rightarrow 2D = 0 \Rightarrow D = 0 \Rightarrow$ skew symmetric determinant of third order has the value zero.

P-2: If any two rows (or columns) of a determinant be interchanged, the value of determinant is changed in sign only. e.g.

$$D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} \text{ \& } D' = \begin{vmatrix} a_2 & b_2 & c_2 \\ a_1 & b_1 & c_1 \\ a_3 & b_3 & c_3 \end{vmatrix} = D'. \quad \text{Then } D' = -D$$

P-3: If a determinant has any two rows (or columns) identical, then its value is zero. e.g.

$$\text{Let } D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} \text{ then it can be verified that } D = 0.$$

P-4: If all the elements of any row (or column) be multiplied by the same number, then the determinant is multiplied by that number.

$$\text{e.g. If } D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} \text{ and } D' = \begin{vmatrix} Ka_1 & Kb_1 & Kc_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

then $D' = KD$

P-5: If each element of any row (or column) can be expressed as a sum of two terms then the determinant can be expressed as the sum of two determinants. e.g.

$$\begin{vmatrix} a_1 + x & b_1 + y & c_1 + z \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} + \begin{vmatrix} x & y & z \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

P-6: The value of a determinant is not altered by adding to the elements of any row (or column) the same

multiples of the corresponding elements of any other row (or column) e.g. Let $D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ and

$$D' = \begin{vmatrix} a_1 + ma_2 & b_1 + mb_2 & c_1 + mc_2 \\ a_2 & b_2 & c_2 \\ a_3 + na_1 & b_3 + nb_1 & c_3 + nc_1 \end{vmatrix}.$$

Then $D' = D$

Note: that while applying this property atleast one row (or column) must remain unchanged.

P-7: If by putting $x = a$ the value of a determinant vanishes then $(x-a)$ is a factor of the determinant.

7. Multiplication of Two Determinants:

$$(i) \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} \times \begin{vmatrix} l_1 & m_1 \\ l_2 & m_2 \end{vmatrix} = \begin{vmatrix} a_1 l_1 + b_1 l_2 & a_1 m_1 + b_1 m_2 \\ a_2 l_1 + b_2 l_2 & a_2 m_1 + b_2 m_2 \end{vmatrix}$$

Similarly two determinants of order three are multiplied.

$$(ii) \text{ If } D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} \neq 0 \text{ then, } D^2 = \begin{vmatrix} A_1 & B_1 & C_1 \\ A_2 & B_2 & C_2 \\ A_3 & B_3 & C_3 \end{vmatrix} \text{ Where } A_i, B_i, C_i \text{ are cofactors}$$

Proof : Consider

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} \times \begin{vmatrix} A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \\ C_1 & C_2 & C_3 \end{vmatrix} = \begin{vmatrix} D & 0 & 0 \\ 0 & D & 0 \\ 0 & 0 & D \end{vmatrix}$$

Note: $a_1 A_2 + b_1 B_2 + c_1 C_2 = 0$ etc.

$$\text{Therfor, } D \times \begin{vmatrix} A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \\ C_1 & C_2 & C_3 \end{vmatrix} = D^3 \Rightarrow \begin{vmatrix} A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \\ C_1 & C_2 & C_3 \end{vmatrix} = D^2$$

$$\text{Or } \begin{vmatrix} A_1 & B_1 & C_1 \\ A_2 & B_2 & C_2 \\ A_3 & B_3 & C_3 \end{vmatrix} = D^2$$

8. System of Linear Equation (In two Variables) :

(i) Consistent Equations: Definite & unique solution. [Intersecting lines]

(ii) Inconsistent Equations: No solution. [Parallel line]

(iii) Dependent equation: Infinite solutions. [Identical lines]

Let $a_1 x + b_1 y + c_1 = 0$ & $a_2 x + b_2 y + c_2 = 0$

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2} \Rightarrow \text{given equations are inconsistent \&}$$

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} \Rightarrow \text{given equations are dependent}$$

9. Cramer's Rule : [Simultaneous Equatons Involving Three Unknowns]

Let, $a_1 x + b_1 y + c_1 z = d_1$ (I)

$a_2 x + b_2 y + c_2 z = d_2$ (II)

then, $x = \frac{D_1}{D}$, $y = \frac{D_2}{D}$, $z = \frac{D_3}{D}$.

$$\text{where } D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}; D_1 = \begin{vmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_2 \\ d_3 & b_3 & c_3 \end{vmatrix}; D_2 = \begin{vmatrix} a_1 & d_1 & c_1 \\ a_2 & d_2 & c_2 \\ a_3 & d_3 & c_3 \end{vmatrix} \& D_3 = \begin{vmatrix} a_1 & b_1 & d_1 \\ a_2 & b_2 & d_2 \\ a_3 & b_3 & d_3 \end{vmatrix}$$

Note :

- (a) If $D \neq 0$ and at least one of $D_1, D_2, \& D_3 \neq 0$, then the given system of equations are consistent and have unique non trivial solution.
- (b) If $D \neq 0 \& D_1 = D_2 = D_3 = 0$, then the given system of equation are consistent and have trivial solution only.
- (c) If $D = D_1 = D_2 = D_3 = 0$, then the given system of equations are consistent and have infinite solutions.
- (d) If $D = 0$ but atleast one of D_1, D_2, D_3 is not zero then the equations are inconsistent and have no solution.

10. If x, y, z are not all zero, the condition for $a_1x + b_1y + c_1z = 0$; $a_2x + b_2y + c_2z = 0$ & $a_3x + b_3y + c_3z = 0$ to be

consistent in x, y, z is that $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0$.

Remember that if a given system of linear equations have **Only Zero** solution for all its variables then the given equations are said to have **TRIVIAL SOLUTION**.



LEVEL-1 (Single Correct Option)**DAY-1**

- If $A = \begin{bmatrix} 8 & 7 & 2 \\ 5 & 8 & 2 \\ 7 & 2 & 8 \end{bmatrix}$, then $tr(A)$ is equal to :
 (a) 8 (b) 49 (c) 24 (d) none of these.
- $A = \begin{bmatrix} 0 & x+2 \\ 2x-3 & 0 \end{bmatrix}$ is skew symmetric, then x is equal to :
 (a) $\frac{1}{3}$ (b) 5 (c) 3 (d) 1.
- If $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, then $A + A'$ is equal to :
 (a) $\begin{bmatrix} 2 & 5 \\ 5 & 8 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$ (c) $\begin{bmatrix} 2 & 4 \\ 6 & 8 \end{bmatrix}$ (d) none of these.
- If $A = [a_{ij}]_{n \times n}$ is a skew symmetric matrix then $tr(A)$ is equal to :
 (a) 0 (if n is even) (b) a_{ii} [if n is odd, where $i = \frac{n+1}{2}$]
 (c) $\sum_{i=1}^n a_{ii}$ (d) none of these.
- If A is a scalar matrix of size $n \times n$ and $tr(A) = 2n$, then A is equal to :
 (a) $2I_n$ (b) $\frac{1}{2}I_n$ (c) I_n^2 (d) none of these
- If $A = \begin{bmatrix} 0 & -1 & 2 \\ 1 & 0 & 3 \\ -2 & -3 & 0 \end{bmatrix}$, then $A + 2A^T$ equals :
 (a) A (b) $-A^T$ (c) A^T (d) 2^2
- The matrix $\begin{bmatrix} 0 & 5 & -7 \\ -5 & 0 & 11 \\ 7 & -11 & 0 \end{bmatrix}$ is known as :
 (a) symmetric matrix (b) diagonal matrix (c) upper triangular matrix (d) skew-symmetric matrix
- A square matrix A can be expressed as $A = \frac{1}{2}(A + A') + \frac{1}{2}(A - A')$, where :
 (a) $\frac{1}{2}(A + A')$ is a skew symmetric matrix of A (b) $\frac{1}{2}(A - A')$ is a symmetric matrix of A
 (c) $\frac{1}{2}(A - A')$ is a skew symmetric matrix of A (d) none of the above
- A square matrix A is called involutory if :
 (a) $A^2 = A$ (b) $A^2 = 0$ (c) $A^2 = I$ (d) none of these.

10. A square matrix A is called idempotent if :
 (a) $A^2 = A$ (b) $A^2 = 0$ (c) $A^2 = I$ (d) none of these.
11. Matrix theory was introduced by :
 (a) Cauchy Riemann (b) Caley Hamilton (c) Cauchy Schwarz (d) Einstein
12. If $A = [a_{ij}]$ is a scalar matrix of order $n \times n$ such that $a_{ij} = k$ for all i , then $|A|$ is equal to :
 (a) nk (b) $n+k$ (c) n^k (d) k^n
13. If A is an orthogonal matrix, then :
 (a) $|A| = 0$ (b) $|A| = \pm 1$ (c) $|A| = \pm 2$ (d) none of these
14. The matrix $\begin{bmatrix} 0 & 5 & 7 \\ -5 & 0 & 11 \\ -7 & -11 & 0 \end{bmatrix}$ is known as :
 (a) symmetric matrix (b) diagonal matrix
 (c) upper triangular matrix (d) skew-symmetric matrix
15. $A = \begin{bmatrix} 0 & 5 & 7 \\ -5 & 0 & 11 \\ 7 & -11 & 0 \end{bmatrix}$ is
 (a) orthogonal (b) $-A$ (c) null matrix (d) skew symmetric
16. If $A = \begin{bmatrix} 4 & x+2 \\ 2x-3 & x+1 \end{bmatrix}$ is symmetric, then $x =$
 (a) 3 (b) 5 (c) 2 (d) 4
17. If $A+B = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$ and $A-B = \begin{bmatrix} -1 & 1 \\ 0 & -1 \end{bmatrix}$, then A is equal to :
 (a) $\begin{bmatrix} 0 & \frac{1}{2} \\ 1 & 0 \end{bmatrix}$ (b) $\begin{bmatrix} \frac{1}{2} & 0 \\ 1 & 0 \end{bmatrix}$ (c) $\begin{bmatrix} 0 & \frac{1}{2} \\ \frac{1}{2} & 0 \end{bmatrix}$ (d) none of these.
18. If $A+B = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$ and $A-2B = \begin{bmatrix} -1 & 1 \\ 0 & -1 \end{bmatrix}$, then A is equal to :
 (a) $\begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} 2/3 & 1/3 \\ 1/3 & 2/3 \end{bmatrix}$ (c) $\begin{bmatrix} 1/3 & 1/3 \\ 2/3 & 1/3 \end{bmatrix}$ (d) none of these
19. If $A = \begin{bmatrix} 1 & -2 & 4 \\ 2 & 3 & 2 \\ 3 & 1 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & -2 & 4 \\ 1 & 3 & 2 \\ -1 & 1 & 5 \end{bmatrix}$ then $A+B$ is
 (a) $\begin{bmatrix} 1 & -2 & 4 \\ 3 & 3 & 2 \\ 2 & 1 & 5 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & -2 & 8 \\ 3 & 3 & 4 \\ 2 & 1 & 10 \end{bmatrix}$ (c) $\begin{bmatrix} 1 & -4 & 8 \\ 3 & 6 & 4 \\ 2 & 2 & 10 \end{bmatrix}$ (d) none of these
20. A square matrix is an orthogonal matrix if :
 (a) $AA' = 0$ (b) $AA' = I$ (c) $A+A' = I$ (d) none of these

21. Additive inverse of the matrix $\begin{bmatrix} 1 & 0 & 2 & -1 \\ 3 & 4 & 8 & 12 \end{bmatrix}$ is
- (a) $\begin{bmatrix} 3 & 4 & 8 & 12 \\ 1 & 0 & 2 & -1 \end{bmatrix}$ (b) $\begin{bmatrix} -3 & -4 & -8 & -12 \\ 1 & 0 & 2 & -1 \end{bmatrix}$
- (c) $\begin{bmatrix} -1 & 0 & -2 & 1 \\ -3 & -4 & -8 & -12 \end{bmatrix}$ (d) not defined as it is not a square matrix.

22. A matrix $A = [a_{ij}]_{n \times n}$ is said to be lower triangular if
- (a) $a_{ij} = 0$ for $i > j$ (b) $a_{ij} = 0$ for $i < j$ (c) $a_{ij} = 0$ for $i \geq j$ (d) $a_{ij} = 0$ for $i \leq j$

23. A matrix $A = [a_{ij}]_{m \times n}$ is said to be row matrix if
- (a) $m = 1$ (b) $n = 1$ (c) $m = n$ (d) $m \times n$

24. If $\begin{bmatrix} x+y & 8 \\ 0 & x-y \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ -1 & 5 \end{bmatrix} + \begin{bmatrix} 3 & 5 \\ 1 & -2 \end{bmatrix}$ then (x, y) equals
- (a) (4, 1) (b) (1, 4) (c) (-4, -1) (d) (1, -4)

25. If $A = \begin{bmatrix} 0 & 2 & 3 \\ 2 & 1 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 7 & 6 & 3 \\ 1 & 4 & 5 \end{bmatrix}$ then $3A - 2B$ is the matrix
- (a) $\begin{bmatrix} -14 & 6 & -3 \\ 4 & 5 & -2 \end{bmatrix}$ (b) $\begin{bmatrix} 14 & -6 & -3 \\ -4 & -5 & -2 \end{bmatrix}$ (c) $\begin{bmatrix} -14 & -6 & 3 \\ 4 & -5 & 2 \end{bmatrix}$ (d) $\begin{bmatrix} 14 & 6 & -3 \\ 4 & 5 & -2 \end{bmatrix}$

26. Which of the following matrix is not symmetric
- (a) $\begin{bmatrix} 4 & -8 \\ -8 & 4 \end{bmatrix}$ (b) $\begin{bmatrix} 5 & -6 & 1 \\ -6 & 3 & 0 \\ 1 & 0 & 8 \end{bmatrix}$ (c) $\begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix}$ (d) $\begin{bmatrix} 5 & 4 & 3 \\ -4 & 0 & 1 \\ -3 & -1 & 0 \end{bmatrix}$

27. Which of the following may be a skew symmetric matrix ($a_i \neq 0$)
- (a) $\begin{bmatrix} a_1 & a_2 \\ a_3 & a_4 \end{bmatrix}$ (b) $\begin{bmatrix} a_1 & 0 & 0 \\ 0 & a_2 & 0 \\ 0 & 0 & a_3 \end{bmatrix}$ (c) $\begin{bmatrix} 0 & a_1 & a_2 \\ a_3 & 0 & a_4 \\ a_5 & a_6 & 0 \end{bmatrix}$ (d) $\begin{bmatrix} a_1 & 1 & 2 \\ -1 & a_2 & 3 \\ -2 & -3 & a_3 \end{bmatrix}$

28. If $A = \begin{bmatrix} 2 & 3 & 1 \\ 4 & -2 & 6 \end{bmatrix}$; $B = \begin{bmatrix} 0 & 3 & 2 \\ -2 & 1 & 4 \end{bmatrix}$ and $C = \begin{bmatrix} 7 & 8 \\ 9 & 0 \\ 6 & 5 \end{bmatrix}$ then $2A + 3B - C^T$ equals.
- (a) $\begin{bmatrix} -3 & 6 & 2 \\ -6 & -1 & 19 \end{bmatrix}$ (b) $\begin{bmatrix} 3 & 6 & 2 \\ 6 & 1 & 19 \end{bmatrix}$ (c) $\begin{bmatrix} 11 & 24 & 14 \\ 10 & 7 & 31 \end{bmatrix}$ (d) none of these

29. The trace of the matrix $\begin{bmatrix} 2 & 0 & 3 \\ 7 & -2 & 3 \\ 1 & 1 & 5 \end{bmatrix}$ is
- (a) 9 (b) 5 (c) 1 (d) -4

30. $\begin{bmatrix} 1 & -2 & 3 \\ 2 & -1 & 4 \\ 3 & 4 & 1 \end{bmatrix}$ is a
 (a) rectangular matrix (b) column matrix (c) square matrix (d) Row Matrix

DAY-2

1. If $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} a & b \\ 0 & a \end{bmatrix}$ are two matrices, then :
 (a) $AB = I$ (b) $AB = BA$ (c) $AB \neq BA$ (d) none of these
2. If I_n is an identity matrix, then (k is a natural number):
 (a) $(I_n)^k = I_n$ (b) $(I_n)^k = kI_n$ (c) $(I_n)^k = k^n I_n$ (d) none of these
3. If A is the diagonal matrix (order 3×3) diagonal (d_1, d_2, d_3) , then $A^n; n \in N$ is :
 (a) $\text{diag}(d_1, d_2, d_3)$ (b) $\text{diag}(nd_1, nd_2, nd_3)$ (c) $\text{diag}(d_1^n, d_2^n, d_3^n)$ (d) none of these.
4. If $A = \begin{bmatrix} 2 & 4 \\ 3 & 1 \end{bmatrix}$, and $A^2 - \lambda A + 4 \begin{bmatrix} 0 & 5 \\ \frac{15}{4} & -\frac{15}{4} \end{bmatrix} = 0$, then λ is equal :
 (a) 2 (b) 4 (c) 8 (d) none of these
5. $\begin{bmatrix} 1 & 2 \\ 5 & 7 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, then :
 (a) $x = 1, y = -1$ (b) $x = 0, y = 1$ (c) $x = -1, y = 1$ (d) $x = 1, y = 0$
6. Given $A = \begin{bmatrix} -1 & 1 \\ -2 & 1 \end{bmatrix}$, which of the following result is true:
 (a) $A^2 = I$ (b) $A^2 = \frac{1}{2}I$ (c) $A^2 = -I$ (d) none of these
7. If $A = [a, b], B = [-b - a]$ and $C = \begin{bmatrix} a \\ -a \end{bmatrix}$, then the correct statement is :
 (a) $A = -A$ (b) $A + B = A - B$ (c) $AC = BC$ (d) $CA = CB$
8. If I_n is an identity matrix of order n , then $(I_n)^{-1}$ is equal to :
 (a) I_n (b) 0 (c) does not exist (d) $-I_n$
9. If $A = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$ and $A^2 = A + B$, then B is equal to :
 (a) $\begin{bmatrix} 5 & 4 \\ -4 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} -5 & -4 \\ 4 & -1 \end{bmatrix}$ (c) $\begin{bmatrix} -5 & -4 \\ -4 & -1 \end{bmatrix}$ (d) none of these
10. If $A = [a_{ij}]_{m \times n}$ and $B = [b_{ij}]_{p \times q}$ and $AB = BA$, then :
 (a) $n = p$ (b) $n = p, m = q$ (c) $m = n = p = q$ (d) $m = q$
11. If $A = \begin{bmatrix} 1 & \frac{1}{2} \\ 0 & 1 \end{bmatrix}$, then A^{64} is :
 (a) $\begin{bmatrix} 1 & 32 \\ 32 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 0 \\ 32 & 1 \end{bmatrix}$ (c) $\begin{bmatrix} 1 & 32 \\ 0 & 1 \end{bmatrix}$ (d) none of these

12. If $f(x) = x^2 + 4x - 5$ and $A = \begin{bmatrix} 1 & 2 \\ 4 & -3 \end{bmatrix}$, then $f(A)$ is equal to :
- (a) $\begin{bmatrix} 0 & -4 \\ 8 & 8 \end{bmatrix}$ (b) $\begin{bmatrix} 2 & 1 \\ 2 & 0 \end{bmatrix}$ (c) $\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$ (d) $\begin{bmatrix} 8 & 4 \\ 8 & 0 \end{bmatrix}$
13. If $A = \begin{bmatrix} i & 0 \\ 0 & i \end{bmatrix}$ and $B = \begin{bmatrix} 0 & -i \\ -i & 0 \end{bmatrix}$, then $(A + B)(A - B)$ is equal to :
- (a) $A^2 - B^2$ (b) $A^2 + B^2$ (c) $A^2 - B^2 + BA + AB$ (d) none of these.
14. If $A = \begin{bmatrix} 1 & -2 \\ x & y \end{bmatrix}$ and $AA' = I$, then x can be :
- (a) 1 (b) 2 (c) 0 (d) none of these.
15. If $A = \begin{bmatrix} \alpha & 0 \\ 1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 5 & 1 \end{bmatrix}$, then value of α for which $A^2 = B$, is :
- (a) 1 (b) -1 (c) 4 (d) no real value
16. If $A = \begin{bmatrix} 1 & 0 \\ 2 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 0 \\ 1 & 12 \end{bmatrix}$, then :
- (a) $|AB| = 0, |BA| = 0$ (b) $AB = BA$ (c) $|AB| \neq 0, |BA| = 0$ (d) $|AB| \neq 0, |BA| \neq 0$
17. If $A = \begin{bmatrix} 2 & 3 \\ 3 & 4 \end{bmatrix}$ and $A^2 - kA - 5I_2 = 0$, then the value of k is :
- (a) 3 (b) 5 (c) 7 (d) -7
18. If A and B are square matrices of equal degree, then which one is correct?
- (a) $A + B = B + A$ (b) $A + B = A - B$ (c) $A - B = B - A$ (d) $AB = BA$
19. If $U = [2 - 3 4]$, $V = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$, $X = [0 2 3]$ and $Y = \begin{bmatrix} 2 \\ 2 \\ 4 \end{bmatrix}$, then $UV + XY$ is equal to :
- (a) 20 (b) [-20] (c) -20 (d) [20]
20. If $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ a & b & -1 \end{bmatrix}$, then A^2 is equal to :
- (a) A (b) $-A$ (c) null matrix (d) I
21. If A and B are two matrices such that $A + B$ and AB are both defined, then :
- (a) A and B are two matrices not necessarily of same order
 (b) A and B are square matrices of same order
 (c) number of columns of A = number of rows of B
 (d) none of the above
22. If $X = \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix}$, then value of X^n is :
- (a) $\begin{bmatrix} 3n & -4n \\ n & -n \end{bmatrix}$ (b) $\begin{bmatrix} 2+n & 5-n \\ n & -n \end{bmatrix}$ (c) $\begin{bmatrix} 3^n & (-4)^n \\ 1^n & (-1)^n \end{bmatrix}$ (d) none of these
23. If $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, then A^4 is equal to :
- (a) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$ (c) $\begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}$ (d) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

24. Let $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, $B = \begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix}$, $a, b, \in \mathbb{N}$. then :
- (a) There cannot exist any B such that $AB = BA$
 (b) There exists more than one but finite number of B such that $AB = BA$
 (c) There exists infinitely many B such that $AB = BA$
 (d) There exists exactly one B such that $AB = BA$
25. If the matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ then A^2 is
- (a) $\begin{bmatrix} a^2 & b^2 \\ c^2 & d^2 \end{bmatrix}$ (b) $\begin{bmatrix} a^2+bc & ab+bd \\ ac+dc & bc+d^2 \end{bmatrix}$ (c) nonexistent (d) none of these
26. If $A = \begin{bmatrix} \alpha & 0 \\ 1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 5 & 1 \end{bmatrix}$ such that $A^2 = B$ then α is
- (a) 1 (b) -1 (c) 4 (d) none of these
27. If $A = \begin{bmatrix} 4 & -1 & -4 \\ 3 & 0 & -4 \\ 3 & -1 & -3 \end{bmatrix}$ then A^2 is equal to
- (a) A (b) I (c) A^T (d) none of these
28. If $A = \begin{bmatrix} 1 & \omega & \omega^2 \\ \omega & \omega^2 & 1 \\ \omega^2 & 1 & \omega \end{bmatrix}$, $B = \begin{bmatrix} \omega & \omega^2 & 1 \\ \omega^2 & 1 & \omega \\ \omega & \omega^2 & 1 \end{bmatrix}$ and $C = \begin{bmatrix} 1 \\ \omega \\ \omega^2 \end{bmatrix}$ where ω is the complex cube root of 1 then $(A+B)C$ is equal to
- (a) $\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ (c) $\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$ (d) $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$
29. If $\begin{bmatrix} 1 & x & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 & 2 \\ 2 & 5 & 1 \\ 15 & 3 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ x \end{bmatrix} = 0$ then x is
- (a) 2 (b) -2 (c) 14 (d) none of these
30. If $\begin{bmatrix} x+y & y \\ 2x & x-y \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$ then $x \cdot y$ is equal to
- (a) -5 (b) 5 (c) 4 (d) 6
31. The product AB of the matrix A and B is possible if
- (a) number of columns of A and B are equal (b) number of rows of A and B are equal
 (c) number of rows in A is equal to the number of columns in B
 (d) number of columns in A is equal to number of rows in B
32. If A is a matrix of order 2×3 and AB is the matrix of order 2×5 , then B may be a
- (a) 3×5 matrix (b) 5×3 matrix (c) 3×2 matrix (d) 5×2 matrix
33. If A is a square matrix of $m \times m$ then order of A^n is
- (a) $n \times n$ (b) $m \times m$ (c) $m \times n$ (d) $n \times m$
34. Which of the following is true, if A and B are square matrices of the same order
- (a) $(A+B)^2 = A^2 + B^2 + 2AB$ (b) $(A+B)(A-B) = A^2 - B^2$
 (c) $(A+B)^2 + (A-B)^2 = 2A^2 + 2B^2$ (d) none of these
35. If $A = \begin{bmatrix} i & 0 \\ 0 & i \end{bmatrix}$ then A^n is equal to ($i = \sqrt{-1}$)
- (a) A for $n = 4$ (b) -A for $n = 6$ (c) -I for $n = 5$ (d) I for $n = 8$

DAY-3

- The value of the determinant $\begin{vmatrix} 1 & 2 & 3 \\ 3 & 5 & 7 \\ 8 & 14 & 20 \end{vmatrix}$ is :
 (a) 20 (b) 10 (c) 0 (d) 5
- The value of determinant $\begin{vmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{vmatrix}$ is equal to :
 (a) -4 (b) 0 (c) 1 (d) 4
- If $x^n y^b = e^m$, $e^c y^d = e^n$, $\Delta_1 = \begin{vmatrix} m & b \\ n & d \end{vmatrix}$, $\Delta_2 = \begin{vmatrix} a & m \\ c & n \end{vmatrix}$ and $\Delta_3 = \begin{vmatrix} a & b \\ c & d \end{vmatrix}$ then value of x and y are respectively :
 (a) $\frac{\Delta_1}{\Delta_3}$ and $\frac{\Delta_2}{\Delta_3}$ (b) $\frac{\Delta_2}{\Delta_1}$ and $\frac{\Delta_3}{\Delta_1}$ (c) $\log\left(\frac{\Delta_1}{\Delta_3}\right)$ and $\log\left(\frac{\Delta_2}{\Delta_3}\right)$ (d) e^{Δ_1/Δ_3} and e^{Δ_2/Δ_3}
- If $C = 2 \cos \theta$ then the value of the determinant $\Delta = \begin{vmatrix} C & 1 & 0 \\ 1 & C & 1 \\ 6 & 1 & C \end{vmatrix}$ is :
 (a) $\frac{\sin 4\theta}{\sin \theta}$ (b) $\frac{2 \sin^2 2\theta}{\sin \theta}$ (c) $4 \cos^2 \theta (2 \cos \theta - 1)$ (d) none of these
- If every element of a third order determinant of value Δ is multiplied by 5, then the value of the new determinant is :
 (a) Δ (b) 5Δ (c) 25Δ (d) 125Δ
- The value of the determinant is $\begin{vmatrix} 1 & 2 & 3 \\ 3 & 5 & 7 \\ 8 & 14 & 20 \end{vmatrix}$:
 (a) 20 (b) 10 (c) 0 (d) 5
- If $x + y + z = \pi$, then the value of $\Delta = \begin{vmatrix} \sin(x+y+z) & \sin B & \cos C \\ -\sin B & 0 & \tan A \\ \cos(A+B) & -\tan A & 0 \end{vmatrix}$ is equal to :
 (a) 0 (b) $2 \sin B \tan A \cos C$ (c) 1 (d) none of these
- The roots of the equation $\begin{vmatrix} x-1 & 1 & 1 \\ 1 & x-1 & 1 \\ 1 & 1 & x-1 \end{vmatrix} = 0$ are :
 (a) 1, 2 (b) -1, 2 (c) 1, -2 (d) -1, -2
- $\begin{vmatrix} 1 & 1 & 1 \\ 1 & 1+x & 1 \\ 1 & 1 & 1+y \end{vmatrix}$ equals :
 (a) $x+y$ (b) xy (c) $x-y$ (d) $1+x+y$
- If $\begin{vmatrix} 1+x & x & x^2 \\ x & 1+x & x^2 \\ x^2 & x & 1+x \end{vmatrix} = a + bx + cx^2 + dx^3 + ex^4 + fx^5$, then a is equal to :
 (a) 0 (b) 1 (c) 2 (d) none of these.
- If $p\lambda^4 + q\lambda^3 + r\lambda^2 + s\lambda + t = \begin{vmatrix} \lambda^2 + 3\lambda & \lambda - 1 & \lambda - 3 \\ \lambda + 1 & 2 - \lambda & \lambda - 3 \\ \lambda - 3 & \lambda + 4 & 3\lambda \end{vmatrix}$, then t equals :
 (a) 33 (b) -39 (c) 27 (d) 24



12. $\begin{vmatrix} 11 & 12 & 13 \\ 12 & 13 & 14 \\ 13 & 14 & 15 \end{vmatrix}$ is equal to :
 (a) 1 (b) 0 (c) -1 (d) 67
13. $\begin{vmatrix} x & 4 & y+z \\ y & 4 & z+x \\ z & 4 & x+y \end{vmatrix}$ is equal to :
 (a) 4 (b) $x+y+z$ (c) xyz (d) 0
14. If $A = \begin{bmatrix} \frac{1}{a} & a^2 & bc \\ \frac{1}{b} & b^2 & ca \\ \frac{1}{c} & c^2 & ab \end{bmatrix}$, then $|A|$ is :
 (a) 0 (b) 1 (c) -1 (d) abc
15. If $z = \begin{vmatrix} 2 & 5-i & 7+i \\ 5+i & 2 & 3-i \\ 7-i & 3+i & 7 \end{vmatrix}$, then :
 (a) $\arg(z)$ is 0 or π (b) $\arg(z)$ is $\frac{\pi}{2}$ or $-\frac{\pi}{2}$ (c) $0 < \arg(z) < \frac{\pi}{2}$ (d) none of these.
16. The determinant $\begin{vmatrix} b+c & a-b & a \\ c+a & b-c & b \\ a+b & c-a & c \end{vmatrix}$ is equal to :
 (a) $\begin{vmatrix} c & b & a \\ a & c & b \\ b & a & c \end{vmatrix}$ (b) $\begin{vmatrix} c & a & b \\ a & b & c \\ b & c & a \end{vmatrix}$ (c) $\begin{vmatrix} a & c & b \\ b & a & c \\ c & b & a \end{vmatrix}$ (d) none of these.
17. $\begin{vmatrix} \sin^2 x & \cos^2 x & 1 \\ \cos^2 x & \sin^2 x & 1 \\ -10 & 12 & 2 \end{vmatrix}$ is equal to :
 (a) 0 (b) $12\cos^2 x - 10\sin^2 x$
 (c) $12\sin^2 x - 10\cos^2 x - 2$ (d) $10\sin 2x$
18. If $x+y+z=\pi$, then the value of $\Delta = \begin{vmatrix} \sin(x+y+z) & \sin B & \cos C \\ -\sin B & 0 & \tan A \\ \cos(A+B) & -\tan A & 0 \end{vmatrix}$ is equal to :
 (a) 0 (b) $2\sin B \tan A \cos C$ (c) 1 (d) none of these
19. $\begin{vmatrix} a+b & b+c & c+a \\ b+c & c+a & a+b \\ c+a & a+b & b+c \end{vmatrix} = k \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$, then k is equal to:
 (a) 1 (b) 2 (c) 3 (d) 4
20. If $\begin{vmatrix} 6i & -3i & 1 \\ 4 & 3i & -1 \\ 20 & 3 & i \end{vmatrix} = x+iy$ then
 (a) $x=3, y=1$ (b) $x=1, y=3$ (c) $x=0, y=3$ (d) $x=0, y=0$

DAY-4

1. If a, b, c are all different and $\begin{vmatrix} a & a^3 & a^4-1 \\ b & b^3 & b^4-1 \\ c & c^3 & c^4-1 \end{vmatrix} = 0$, then the value of $abc(ab+bc+ca)$ is :
 (a) $a+b+c$ (b) 0 (c) $a^2+b^2+c^2$ (d) $a^2-b^2+c^2$
2. If $\begin{vmatrix} -a^2 & ab & ac \\ ab & -b^2 & bc \\ ac & bc & -c^2 \end{vmatrix} = \lambda a^2 b^2 c^2$, then the value of λ is :
 (a) 2 (b) -2 (c) 4 (d) 55
3. The value of the determinant $\begin{vmatrix} x+1 & x+2 & x+4 \\ x+3 & x+5 & x+8 \\ x+7 & x+10 & x+14 \end{vmatrix}$ is :
 (a) -2 (b) x^2+2 (c) 2 (d) 0
4. The value of the determinant $\begin{vmatrix} 1 & \cos(\alpha-\beta) & \cos\alpha \\ \cos(\alpha-\beta) & 1 & \cos\beta \\ \cos\alpha & \cos\beta & 1 \end{vmatrix}$ is :
 (a) 0 (b) 1 (c) $\alpha^2-\beta^2$ (d) $\alpha^2+\beta^2$
5. If $1, \omega, \omega^2$ are the cube roots of unity, then $\Delta = \begin{vmatrix} 1 & \omega^n & \omega^{2n} \\ \omega^n & \omega^{2n} & 1 \\ \omega^{2n} & 1 & \omega^n \end{vmatrix}$ is equal to :
 (a) 0 (b) 1 (c) ω (d) ω^2
6. A root of the equation $\begin{vmatrix} 3x-8 & 3 & 3 \\ 3 & 3x-8 & 3 \\ 3 & 3 & 3x-8 \end{vmatrix} = 0$ is :
 (a) $8/3$ (b) $2/3$ (c) $1/3$ (d) none
7. The roots of equation $\begin{vmatrix} x-1 & 1 & 1 \\ 1 & x-1 & 1 \\ 1 & 1 & x-1 \end{vmatrix} = 0$ are :
 (a) 1, 2 (b) -1, 2 (c) 1, -2 (d) -1, -2
8. If one root of the equation $\begin{vmatrix} 7 & 6 & x \\ 2 & x & 2 \\ x & 3 & 7 \end{vmatrix} = 0$ is $x = -9$, then the other roots are :
 (a) 2, 6 (b) 3, 6 (c) 2, 7 (d) 3, 7
9. If $\begin{vmatrix} x+a & b & c \\ a & x+b & c \\ a & b & x+c \end{vmatrix} = 0$, then x equals
 (a) $a+b+c$ (b) $-(a+b+c)$ (c) 0, $a+b+c$ (d) 0, $-(a+b+c)$
10. Sum of the roots of the equation $\begin{vmatrix} 3-x & -6 & 3 \\ -6 & 3-x & 3 \\ 3 & 3 & -6-x \end{vmatrix} = 0$ is :
 (a) 6 (b) 3 (c) 10 (d) none



11. The determinant $\begin{vmatrix} xp+y & x & y \\ yp+z & y & z \\ 0 & xp+y & yp+z \end{vmatrix} = 0$, if :
- (a) x, y, z are in A.P. (b) x, y, z are in G.P. (c) x, y, z are in H.P. (d) xy, yz, zx are in A.P.
12. The value of the determinant $\begin{vmatrix} 1+a & 1 & 1 \\ 1 & 1+a & 1 \\ 1 & 1 & 1+a \end{vmatrix}$ is zero, if :
- (a) $a = -3$ (b) $a = 0$ (c) $a = 2$ (d) (a) or (b)
13. If ω is a cube root of unity, then a root of the equation $\begin{vmatrix} x+1 & \omega & \omega^2 \\ \omega & x+\omega^2 & 1 \\ \omega^2 & 1 & x+\omega \end{vmatrix} = 0$ is :
- (a) $x = 1$ (b) $x = \omega$ (c) $x = \omega^2$ (d) $x = 0$
14. If $a \neq b \neq c$, one value of x which satisfy the equation $\begin{vmatrix} 0 & x-a & x-b \\ x+a & 0 & x-c \\ x+b & x+c & 0 \end{vmatrix} = 0$ is given by :
- (a) $x = a$ (b) $x = b$ (c) $x = c$ (d) $x = 0$
15. Solution of the equation $\begin{vmatrix} a^2 & a & 1 \\ \sin(n+1)x & \sin nx & \sin(n-1)x \\ \cos(n+1)x & \cos nx & \cos(n-1)x \end{vmatrix} = 0$:
- (a) $n\pi/2$ (b) $n\pi$ (c) $2n\pi$ (d) none
16. If $\begin{vmatrix} x & x+y & x+y+z \\ 2x & 3x+2y & 4x+3y+2z \\ 3x & 6x+3y & 10x+6y+3z \end{vmatrix} = 64$, then the real value of x is :
- (a) 2 (b) 3 (c) 4 (d) 6
17. If $\begin{vmatrix} a+x & a-x & a-x \\ a-x & a+x & a-x \\ a-x & a-x & a+x \end{vmatrix} = 0$, then x is equal to :
- (a) 0, $2a$ (b) $a, 2a$ (c) 0, $3a$ (d) none
18. The value of x for which $\begin{vmatrix} x-\omega^2 & \omega & 1 \\ \omega & \omega^2 & 1+x \\ 1 & x+\omega & \omega^2 \end{vmatrix} = 0$ is
- (a) $x = 0$ (b) $x = 1$ (c) $x = -1$ (d) none
19. $\begin{vmatrix} x+1 & \omega & \omega^2 \\ x+\omega & \omega^2 & 1 \\ x+\omega^2 & 1 & \omega \end{vmatrix} = 3$ is an equation of x , where ω, ω^2 are the imaginary cube roots of unity, then the value of x is
- (a) 0 (b) 1 (c) -1 (d) none
20. $\begin{vmatrix} 0 & x-a & x-b \\ x+a & 0 & x-c \\ x+b & x+c & 0 \end{vmatrix} = 0$ is given by :
- (a) $x = a$ (b) $x = b$ (c) $x = -1$ (d) none

21. The determinant $\begin{vmatrix} a & b & a-b \\ b & c & b-c \\ 2 & 1 & 0 \end{vmatrix}$ is equal to zero if a, b, c are in :
- (a) G.P. (b) A.P. (c) H.P. (d) none
22. If $a^2 + b^2 + c^2 = -2$ and $f(x) = \begin{vmatrix} 1+a^2x & (1+b^2)x & (1+c^2)x \\ b & 1+b^2x & (1+c^2)x \\ 2 & (1+b^2)x & 1+c^2x \end{vmatrix}$, then $f(x)$ is a polynomial of degree :
- (a) 1 (b) 0 (c) 3 (d) 2
23. If $\begin{vmatrix} a & a^2 & 1+a^3 \\ b & b^2 & 1+b^3 \\ c & c^2 & 1+c^3 \end{vmatrix} = 0$ and vectors $(1, a, a^2), (1, b, b^2)$ and $(1, c, c^2)$ are non-coplanar, then the product abc equals :
- (a) 2 (b) -1 (c) 1 (d) 0
24. The number of distinct real roots of $\begin{vmatrix} \sin x & \cos x & \cos x \\ \cos x & \sin x & \cos x \\ \cos x & \cos x & \sin x \end{vmatrix} = 0$ in the interval $-\frac{\pi}{4} \leq x \leq \frac{\pi}{4}$ is :
- (a) 0 (b) 2 (c) 1 (d) 3
25. The equation $\begin{vmatrix} x-a & x-b & x-c \\ x-b & x-c & x-a \\ x-c & x-a & x-b \end{vmatrix} = 0$, where a, b, c are different is satisfied by :
- (a) $x = 0$ (b) $x = a$ (c) $x = \frac{1}{3}(a+b+c)$ (d) $x = a+b+c$
26. The value of the determinant $\begin{vmatrix} x & x+a & x+2a \\ x+1 & x+2a & x+4a \\ x+2 & x+3a & x+6a \end{vmatrix}$ is :
- (a) 0 (b) $a^3 - x^3$ (c) $x^3 - a^3$ (d) $(x-a)^3$
27. Given that $abc = -1$, then the value of the determinant $\begin{vmatrix} a & a^2 & 1+a^3 \\ b & b^2 & 1+b^3 \\ c & c^2 & 1+c^3 \end{vmatrix}$ is :
- (a) abc (b) $2abc$ (c) positive (d) 0
28. If $1, \alpha, \alpha^2, \dots, \alpha^{n-1}$ are n th root of unity, then the value of determinant $\begin{vmatrix} \alpha^n & \alpha^{2n} & 1 \\ \alpha^{2n} & 1 & \alpha^n \\ 1 & \alpha^n & \alpha^{2n} \end{vmatrix}$ is equal to :
- (a) 1 (b) -1 (c) 0 (d) none of these
29. If $a_1, a_2, a_3, \dots, a_n$ are in G.P., then $\begin{vmatrix} \log a_1 & \log a_2 & \log a_3 \\ \log a_4 & \log a_5 & \log a_6 \\ \log a_7 & \log a_8 & \log a_9 \end{vmatrix}$ is equal to :
- (a) 1 (b) 0 (c) 9 (d) none of these.
30. If $2^{a_1}, 2^{a_2}, 2^{a_3}, \dots, 2^{a_n}$ are in G.P., then $\begin{vmatrix} a_1 & a_2 & a_3 \\ a_{n+1} & a_{n+2} & a_{n+3} \\ a_{2n+1} & a_{2n+2} & a_{2n+3} \end{vmatrix}$ is equal to :
- (a) 2 (b) 2^3 (c) 0 (d) none of these.

31. If $\begin{vmatrix} a^n & a^{n+1} & a^{n+2} \\ b^n & b^{n+1} & b^{n+2} \\ c^n & c^{n+1} & c^{n+2} \end{vmatrix} = (a-b)(b-c)(c-a)$, then n is equal to :
 (a) $n=1$ (b) $n=2$ (c) any value of n (d) none of these
32. If a, b, c are in A.P., then the value of $\begin{vmatrix} 1 & 2 & a \\ 2 & 3 & b \\ 3 & 4 & c \end{vmatrix}$ is :
 (a) $a+c$ (b) $2(a+c)$ (c) b (d) none of these.
33. If $\begin{vmatrix} 1 & 2 & \log a_1 \\ 2 & 3 & \log a_2 \\ 3 & 4 & \log a_3 \end{vmatrix} = 0$, then :
 (a) a_1, a_2, a_3 are in A.P. (b) a_1, a_2, a_3 are in G.P.
 (c) a_1, a_2, a_3 are in H.P. (d) none of the above.
34. $\begin{vmatrix} 1+x & 1 & 1 \\ 1 & 1+y & 1 \\ 1 & 1 & 1+z \end{vmatrix}$ equals :
 (a) $x+y+z$ (b) $xyz+xy+yz+zx$ (c) $x^3+y^3+z^3$ (d) $x^2y^2z^2$
35. If $\begin{vmatrix} x+1 & 2 & 3 \\ 2 & x+2 & 3 \\ 2 & 2 & x+3 \end{vmatrix} = 0$, then one value of x is :
 (a) 1 (b) -1 (c) 2 (d) none of these.
36. If ω is a cube root of unity, then the root of the following equation $\begin{vmatrix} x+2 & \omega & \omega^2 \\ \omega & x+1+\omega^2 & 1 \\ \omega^2 & 1 & x+1+\omega \end{vmatrix} = 0$ is :
 (a) 0 (b) ω (c) ω^2 (d) -1
37. A root of the equation $\begin{vmatrix} 3-x & -6 & 3 \\ -6 & 3-x & 3 \\ 3 & 3 & -6-x \end{vmatrix} = 0$ is :
 (a) 6 (b) 3 (c) 0 (d) none of these.
38. The value of $\begin{vmatrix} 1+\alpha & 1+\alpha x & 1+\alpha x^2 \\ 1+\beta & 1+\beta x & 1+\beta x^2 \\ 1+\gamma & 1+\gamma x & 1+\gamma x^2 \end{vmatrix}$ depends on :
 (a) α, β, γ (b) x only (c) α, β (d) none of these.
39. $\begin{vmatrix} 1 & a & a^2-bc \\ 1 & b & b^2-ca \\ 1 & c & c^2-ab \end{vmatrix}$ is equal to :
 (a) 1 (b) abc (c) 0 (d) none of these.
40. $\begin{vmatrix} x & p & q \\ p & x & q \\ p & q & x \end{vmatrix}$ is equal to :
 (a) $(x+p)(x+q)(x-p-q)$ (b) $(x-p)(x-q)(x+p+q)$
 (c) $(x-p)(x-q)(x-p-q)$ (d) $(x+p)(x+q)(x+p+q)$

DAY-5

1. If $A = \begin{bmatrix} \cos x & \sin x \\ \sin x & -\cos x \end{bmatrix}$ and $A(\text{adj}A) = \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, then λ is equal to :
 (a) $\sin x \cos x$ (b) -1 (c) -2 (d) none of these.
2. If $AB = I$ and $B = A'$, then :
 (a) $A^{-1} = A'$ (b) $A^{-1} = A$ (c) $A^{-1} = A^2$ (d) none of these
3. If $1, \omega, \omega^2$ are cube roots of unity, then which of the following matrices exists its inverse:
 (a) $\begin{bmatrix} \omega^2 & \omega \\ \omega^4 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} \omega^3 & \omega^4 \\ \omega & 1 \end{bmatrix}$ (c) $\begin{bmatrix} \omega^{2n} & \omega \\ \omega^2 & \omega^n \end{bmatrix}$ (d) none of these
4. If $A = \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}$, then A^{-1} is equal to :
 (a) $\begin{bmatrix} 5 & 2 \\ 3 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} -5 & 3 \\ 2 & -1 \end{bmatrix}$ (c) $\begin{bmatrix} -5 & 2 \\ 3 & -1 \end{bmatrix}$ (d) none of these.
5. $|\text{adj}A|$ is equal to :
 (a) $|A|$ (b) $|A|^{-1}$ (c) $|A|^{n-1}$ (d) none of these
6. If A and B are square matrices of the same order, then which of the following statements cannot be true:
 (a) $(A+B)^{-1} = B^{-1} + A^{-1}$ (b) $(AB)^{-1} = B^{-1}A^{-1}$
 (c) $AB=0 \Rightarrow |A|=0$ or $|B|=0$ (d) $\text{adj}(AB) = (\text{adj}B)(\text{adj}A)$
7. If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, then $\text{adj}(\text{adj}A)$ is equal to :
 (a) $\text{adj}A$ (b) A (c) A' (d) $-A$
8. If A satisfies the equation $x^3 - 5x^2 + 4x + k = 0$, then A^{-1} exists if :
 (a) $k \neq 1$ (b) $k \neq 2$ (c) $k \neq -1$ (d) none of these.
9. If $A = \begin{bmatrix} x & 1 \\ 1 & 0 \end{bmatrix}$ and $A^2 = I$, then A^{-1} is equal to :
 (a) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ (c) $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ (d) $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$
10. If $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, then $\text{adj}(A)$ is equal to :
 (a) $\begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix}$ (b) $-\frac{1}{2} \begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix}$ (c) $\begin{bmatrix} 4 & 2 \\ 3 & 1 \end{bmatrix}$ (d) none of these.
11. If the matrix A is such that $\begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix} \cdot A = \begin{bmatrix} -4 & 1 \\ 7 & 7 \end{bmatrix}$, then A is equal to :
 (a) $\begin{bmatrix} -1 & 3 \\ 2 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} -1 & -2 \\ -3 & -1 \end{bmatrix}$ (c) $\begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix}$ (d) $\begin{bmatrix} -1 & 2 \\ 3 & 1 \end{bmatrix}$
12. A square, non-singular matrix A satisfies $A^2 - A + 2I = 0$, then A^{-1} is equal to :
 (a) $I - A$ (b) $\frac{1}{2}(I - A)$ (c) $\frac{1}{2}(I + A)$ (d) $I + A$

13. The inverse of matrix $A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ is :
- (a) A (b) A^T (c) I (d) $\begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$
14. Which of the following is wrong (A is a matrix $n \times n$)?
- (a) $(\det A^{-1}) = (\det A)^{-1}$ (b) $\det(kA) = k^n (\det A)$
(c) $\det(kA) = k(\det A)$ (d) $A + A + A = 3A$
15. Which of the following is an invertible matrix?
- (a) $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ (b) $\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$ (c) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ (d) $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$
16. If $A = \begin{bmatrix} 1 & -1 & 1 \\ 0 & 2 & -3 \\ 2 & 1 & 0 \end{bmatrix}$, $B = \text{Adj}(A)$ and $C = 5A$ then $\frac{|\text{Adj}(B)|}{|C|}$ is equal to
- (a) 5 (b) 25 (c) -1 (d) 1
17. If $A = \begin{bmatrix} 1 & 2 & -1 \\ -1 & 1 & 2 \\ 2 & -1 & 1 \end{bmatrix}$ then $\text{Det.} [\text{adj}(\text{adj}(A))]$ is
- (a) 12^4 (b) 13^4 (c) 14^4 (d) none of these
18. If the matrix $A, B, (A+B)$ are non-singular, then $(A(A+B)^{-1}B)^{-1}$ is :
- (a) $A+B$ (b) $A^{-1} + B^{-1}$ (c) $(A+2B)^{-1}$ (d) none of these
19. The matrix $\begin{bmatrix} 5 & 10 & 3 \\ 2 & -4 & 6 \\ -1 & -2 & b \end{bmatrix}$ is a singular matrix, if b is equal to :
- (a) -3 (b) 3 (c) 0 (d) for any value of b
20. For how many value of x in the closed interval $[-4, -1]$, is the matrix $\begin{bmatrix} 3 & -1+x & 2 \\ 3 & -1 & x+2 \\ x+3 & -1 & 2 \end{bmatrix}$ non-singular ?
- (a) 2 (b) 0 (c) Finite (d) Infinite
21. Let $X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$, $A = \begin{bmatrix} 1 & -1 & 2 \\ 2 & 0 & 1 \\ 3 & 2 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 3 \\ 1 \\ 4 \end{bmatrix}$. If $AX = B$, then X is equal to :
- (a) $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ (b) $\begin{bmatrix} -1 \\ -2 \\ 3 \end{bmatrix}$ (c) $\begin{bmatrix} -1 \\ -2 \\ -3 \end{bmatrix}$ (d) $\begin{bmatrix} -1 \\ 2 \\ 3 \end{bmatrix}$
22. Let $F(\alpha) = \begin{bmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$, where $\alpha \in \mathbb{R}$, then $[F(\alpha)]^{-1}$ is equal to :
- (a) $F(-\alpha)$ (b) $F(\alpha^{-1})$ (c) $F(2\alpha)$ (d) none of these
23. If $A \neq 0$ and $B \neq 0$ are $n \times n$ matrix such that $AB = 0$, then :
- (a) $\det(A) = 0$ or $\det(B) = 0$ (b) $\det(A) = 0$ and $\det(B) = 0$
(c) $\det(A) = \det(B) \neq 0$ (d) $A^{-1} = B^{-1}$

24. If matrix $A = \begin{bmatrix} 3 & 2 & 4 \\ 1 & 2 & -1 \\ 0 & 1 & 1 \end{bmatrix}$ and $A^{-1} = \frac{1}{k} \text{adj } A$, then k is :
 (a) 7 (b) -7 (c) 1/7 (d) 11
25. Matrix $A = \begin{bmatrix} 1 & 0 & -k \\ 2 & 1 & 3 \\ k & 0 & 1 \end{bmatrix}$ is invertible for :
 (a) $k = 1$ (b) $k = -1$ (c) all real k (d) none of these
26. If matrix $A = \begin{bmatrix} 1 & -1 & -1 \\ 2 & 1 & -3 \\ 1 & 1 & 1 \end{bmatrix}$ and $10B = \begin{bmatrix} 4 & 2 & 2 \\ -5 & 0 & \alpha \\ 1 & -2 & 3 \end{bmatrix}$, if B is the inverse of matrix A , then α is :
 (a) -2 (b) 1 (c) 2 (d) 5
27. If I_3 is identity matrix of order 3, then $(I_3)^{-1} =$
 (a) 0 (b) $3 I_3$ (c) I_3 (d) Not necessarily exists
28. If $A = \begin{bmatrix} 3 & 2 \\ 0 & 1 \end{bmatrix}$, then $(A^{-1})^3$ is equal to :
 (a) $\frac{1}{27} \begin{bmatrix} 1 & -26 \\ 0 & 27 \end{bmatrix}$ (b) $\frac{1}{27} \begin{bmatrix} 1 & 26 \\ 0 & 27 \end{bmatrix}$ (c) $\frac{1}{27} \begin{bmatrix} 1 & -26 \\ 0 & -27 \end{bmatrix}$ (d) $\frac{1}{27} \begin{bmatrix} -1 & -26 \\ 0 & -27 \end{bmatrix}$
29. If $A^2 - A + I = 0$, then the inverse of A is :
 (a) $A + I$ (b) A (c) $A - I$ (d) $I - A$
30. If $A = \begin{bmatrix} 0 & 3 \\ 2 & 0 \end{bmatrix}$ and $A^{-1} = \lambda(A)$, then λ is equal to
 (a) -1/6 (b) 1/3 (c) -1/3 (d) 1/6
31. $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & -2 & 4 \end{bmatrix}$, $I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $6A^{-1} = A^2 + cA + dI$, then c and d respectively equal to :
 (a) -11, 6 (b) -6, 11 (c) 6, 11 (d) -6, -11
32. If $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ satisfies the equation $x^2 - (a+d)x + k = 0$, then A^{-1} exists if
 (a) $k = bc$ (b) $k = ad$ (c) $k = a^2 + b^2 + c^2 + d^2$ (d) none of these
33. If A and B are square matrices of the same order, then which of the following statements cannot be true.
 (a) $(A+B)^{-1} = B^{-1} + A^{-1}$ (b) $(AB)^{-1} = B^{-1}A^{-1}$
 (c) $AB = 0 \Rightarrow |A| = 0$ or $|B| = 0$ (d) $\text{Adj}(AB) = (\text{adj } B)(\text{adj } A)$
34. Ifs N is 2×2 matrix such that $N \begin{pmatrix} -7 & 4 \\ 5 & -8 \end{pmatrix} = \begin{pmatrix} 36 & 0 \\ 0 & 36 \end{pmatrix}$, then $N =$
 (a) $\begin{pmatrix} -8 & -4 \\ 5 & -7 \end{pmatrix}$ (b) $\begin{pmatrix} 8 & 4 \\ 5 & 7 \end{pmatrix}$ (c) $\begin{pmatrix} -8 & -4 \\ -5 & -7 \end{pmatrix}$ (d) none of these
35. If A is a square matrix, then $\text{Adj}(A^T) - (\text{adj } A)^T$ is equal to :
 (a) $2 |A|$ (b) $2 |A| I$ (c) null matrix (d) unit matrix
36. Let $A = \begin{pmatrix} 0 & -\tan \alpha \\ \tan \alpha & 0 \end{pmatrix}$, $B = \begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix}$ and $(I + A)(I - A)^{-1} = B^k$, then k is
 (a) 1 (b) 2 (c) 3 (d) 4

37. Let $A_\alpha = \begin{bmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$, then
- (a) $A_{\alpha+\beta} = A_\alpha A_\beta$ (b) $A_\alpha^{-1} = A_{-\alpha}$ (c) $A_\alpha^{-1} = -A_\alpha$ (d) $A_\alpha^2 = -1$

38. $\begin{bmatrix} 1 & 2 & 4 \\ 2 & 0 & -1 \\ a & -3 & 2 \end{bmatrix}$ will be a singular matrix if a equals

- (a) $\frac{35}{2}$ (b) $\frac{-35}{2}$ (c) 17 (d) -17

39. The cofactor of 6 in the determinant of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$ is

- (a) 3 (b) -3 (c) 6 (d) -6

40. If $A = \begin{bmatrix} 3-2i & 3+5i \\ 2 & 3-2i \end{bmatrix}$, then $\text{adj}A$ is

- (a) $\begin{bmatrix} 3-2i & 2 \\ 3+5i & 3-2i \end{bmatrix}$ (b) $\begin{bmatrix} 3-2i & -2 \\ -3-5i & 3-2i \end{bmatrix}$ (c) $\begin{bmatrix} 3-2i & -3-5i \\ -2 & 3-2i \end{bmatrix}$ (d) $\begin{bmatrix} 3-2i & 3+5i \\ -2 & -3+2i \end{bmatrix}$

DAY-6

1. If $\begin{vmatrix} \alpha & x & x & x \\ x & \beta & x & x \\ x & x & \gamma & x \\ x & x & x & \delta \end{vmatrix} = f(x) - xf'(x)$, then $f(x)$ is

- (a) $(x-\alpha)(x-\beta)(x-\gamma)(x-\delta)$ (b) $(x+\alpha)(x+\beta)(x+\gamma)(x+\delta)$
(c) $2(x-\alpha)(x-\beta)(x-\gamma)(x-\delta)$ (d) none of these

2. Let $f(x) = \begin{vmatrix} x & 1 & 1 \\ \sin \pi x & 2x^2 & 1 \\ x^3 & 3x^4 & 1 \end{vmatrix}$. If $f(x)$ be an odd function and its odd value is equal to $g(x)$ then $f(1)g(1)$ is

- (a) -1 (b) -4 (c) -5 (d) 1

3. If $f(x), g(x)$ and $h(x)$ are three polynomials of degree two and $\varphi(x) = \begin{vmatrix} f(x) & g(x) & h(x) \\ f'(x) & g'(x) & h'(x) \\ f''(x) & g''(x) & h''(x) \end{vmatrix}$, then $\varphi'(x)$ is

- (a) a one degree polynomial (b) a three degree polynomial
(c) a two degree polynomial (d) none of the above.

4. If $\Delta(x) = \begin{vmatrix} x & 1+x^2 & x^3 \\ \log(1+x^2) & e^x & \sin x \\ \cos x & \tan x & \sin^2 x \end{vmatrix}$ then :

- (a) $\Delta(x)$ is divisible by x (b) $\Delta(x) = 0$ (c) $\Delta'(x) = 0$ (d) none of these.

5. If $f(x) = \begin{vmatrix} x & \cos x & e^{x^2} \\ \sin x & x^2 & \sec x \\ \tan x & 1 & 2 \end{vmatrix}$, then the value of $\int_{-\pi/2}^{\pi/2} f(x)dx$ is equal to :

- (a) 0 (b) 1 (c) 2 (d) none of these

6. Let $f(x) = \begin{vmatrix} x^3 & \sin x & \cos x \\ 6 & -1 & 0 \\ p & p^2 & p^3 \end{vmatrix}$ where p is a constant. Then $\frac{d^3}{dx^3}\{f(x)\}$ at $x=0$ is :
- (a) p (b) $p+p^2$ (c) $p+p^3$ (d) independent of p
7. If $\begin{vmatrix} (1+x) & (1+x)^2 & (1+x)^3 \\ (1+x)^4 & (1+x)^5 & (1+x)^6 \\ (1+x)^7 & (1+x)^8 & (1+x)^9 \end{vmatrix} = a_0 + a_1x + a_2x^2 + \dots$, then, a_1 is equal to :
- (a) 1 (b) 2 (c) 0 (d) none of these.
8. If $\Delta_1 = \begin{vmatrix} x & b & b \\ a & x & b \\ a & a & x \end{vmatrix}$ and $\Delta_2 = \begin{vmatrix} x & b \\ a & x \end{vmatrix}$, then :
- (a) $\Delta_1 = 3(\Delta_2)^2$ (b) $\frac{d}{dx}(\Delta_1) = 3\Delta_2$ (c) $\frac{d}{dx}(\Delta_1) = 3\Delta_2^2$ (d) $\Delta_1 = 3(\Delta_2)^{3/2}$
9. If z is a complex number and $a_1, a_2, a_3, b_1, b_2, b_3$ are all real, then $\begin{vmatrix} a_1 + b_1\bar{z} & a_2z + b_2\bar{z} & a_3z + b_3\bar{z} \\ b_1z + a_1\bar{z} & b_2z + a_2\bar{z} & b_3z + a_3\bar{z} \\ b_1z + a_1 & b_2z + a_2 & b_3z + a_3 \end{vmatrix}$ is equal to
- (a) $(a_1a_2a_3 + b_1b_2b_3)^2 |z|^2$ (b) $|z|^2$ (c) 3 (d) 0
10. If a, b, c are in A.P. and $f(x) = \begin{vmatrix} x+a & x^2+1 & 1 \\ x+b & 2x^2-1 & 1 \\ x+c & 3x^2-2 & 1 \end{vmatrix}$, then $f'(x)$ is
- (a) 0 (b) 1 (c) $a+bc$ (d) $\frac{abc}{a+b+c}$
11. If $f(x)$ satisfies the equation $\begin{vmatrix} f(x-3) & f(x+4) & f[(x+1)(x-2)-(x-1)^2] \\ 5 & 4 & -5 \\ 5 & 6 & 15 \end{vmatrix} = 0$ for all real x , then :
- (a) $f(x)$ is not periodic (b) $f(x)$ is periodic and is of period 1
(c) $f(x)$ is periodic and is of period 7 (d) $f(x)$ is an odd function
12. Let $\begin{vmatrix} x & 2 & x \\ x^2 & x & 6 \\ x & x & 6 \end{vmatrix} = Ax^4 + Bx^3 + Cx^2 + Dx + E$. Then the value of $5A + 4B + 3C + 2D + E$ is equal to
- (a) zero (b) -16 (c) 16 (d) -17
13. If $f(x) = \begin{vmatrix} \sin x & \sec x & x^2-1 \\ \operatorname{cosec} x & x \sin x & \cos x \\ \tan x & x \tan x & x^2+1 \end{vmatrix}$ then $\int_{-\pi/3}^{\pi/3} f(x) dx =$
- (a) 1 (b) $\frac{\pi}{3} + 1$ (c) 0 (d) $\frac{\pi}{3} - 1$
14. If α, β are the roots of the equation $x^2 + bx + c = 0$, then $\begin{vmatrix} 3 & 1+\alpha+\beta & 1+\alpha^2+\beta^2 \\ 1+\alpha+\beta & 1+\alpha^2+\beta^2 & 1+\alpha^3+\beta^3 \\ 1+\alpha^2+\beta^2 & 1+\alpha^3+\beta^3 & 1+\alpha^4+\beta^4 \end{vmatrix}$ is equal to
- (a) $(b+1)^2(b^2-4c)$ (b) $(1+b+c)^2(b^2-4c)$
(c) $(b-c)^2(b^2-4c)$ (d) $(1-b-c)^2(b^2-4c)$

15. The value of $\begin{vmatrix} 1 & lm+np & l^2m^2+n^2p^2 \\ 1 & ln+mp & l^2n^2+m^2p^2 \\ 1 & lp+mn & l^2p^2+m^2n^2 \end{vmatrix}$ is
 (a) $(l-p)(m-n)(l-m)(n-p)/(p-m)(l-n)$ (b) 0
 (c) $4(lm+np)(ln+mp)(lp+mn)$ (d) none of these
16. Let $\Delta = \begin{vmatrix} \frac{1-a_1^3b_1^3}{1-a_1b_1} & \frac{1-a_1^3b_2^3}{1-a_1b_2} & \frac{1-a_1^3b_3^3}{1-a_1b_3} \\ \frac{1-a_2^3b_1^3}{1-a_2b_1} & \frac{1-a_2^3b_2^3}{1-a_2b_2} & \frac{1-a_2^3b_3^3}{1-a_2b_3} \\ \frac{1-a_3^3b_1^3}{1-a_3b_1} & \frac{1-a_3^3b_2^3}{1-a_3b_2} & \frac{1-a_3^3b_3^3}{1-a_3b_3} \end{vmatrix}$ then
 (a) $\Delta = 0$ (b) Δ depends only on a_1, a_2, a_3
 (c) Δ depends only on b_1, b_2, b_3 (d) Δ depends on $a_1, a_2, a_3, b_1, b_2, b_3$.
17. If α, β, γ are real numbers, then $\Delta = \begin{vmatrix} 1 & \cos(\beta-\alpha) & \cos(\gamma-\alpha) \\ \cos(\alpha-\beta) & 1 & \cos(\gamma-\beta) \\ \cos(\alpha-\gamma) & \cos(\beta-\gamma) & 1 \end{vmatrix}$ is equal to
 (a) -1 (b) $\cos \alpha \cos \beta \cos \gamma$ (c) $\cos \alpha + \cos \beta + \cos \gamma$ (d) 0
18. Let $\Delta = \begin{vmatrix} 2bc-a^2 & c^2 & b^2 \\ c^2 & 2ca-b^2 & a^2 \\ b^2 & a^2 & 2ab-c^2 \end{vmatrix}$, then Δ can be expressed as
 (a) $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}^2$ (b) $\begin{vmatrix} c & b & a \\ a & b & c \\ c & a & b \end{vmatrix}^2$ (c) $\begin{vmatrix} a & b & c \\ c & b & a \\ c & a & b \end{vmatrix}^2$ (d) none
19. If $\sum \cos^2 \alpha_1 = \sum \cos^2 \beta_1 = \sum \cos^2 \gamma_1 = 1$; $\sum \cos \alpha_1 \cos \beta_1 = \sum \cos \beta_1 \cos \gamma_1 = \sum \cos \gamma_1 \cos \alpha_1 = 0$
 Then the value of $\begin{vmatrix} \cos \alpha_1 & \cos \alpha_2 & \cos \alpha_3 \\ \cos \beta_1 & \cos \beta_2 & \cos \beta_3 \\ \cos \gamma_1 & \cos \gamma_2 & \cos \gamma_3 \end{vmatrix}^2$ is
 (a) 1 (b) -1 (c) 0 (d) none
20. If $u = ax + by + cz, v = ay + bz + cx, w = az + bx + cy$, then the value of $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} \times \begin{vmatrix} x & y & z \\ y & z & x \\ z & x & y \end{vmatrix}$ is
 (a) $u^2 + v^2 + w^2 - 2uvw$ (b) $u^3 + v^3 + w^3 - 3uvw$ (c) 0 (d) none of these
21. If $l_1, m_1, n_1; l_2, m_2, n_2; l_3, m_3, n_3$ are directions cosines of three mutually perpendicular lines, then the value of $\begin{vmatrix} l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \\ l_3 & m_3 & n_3 \end{vmatrix}$ is
 (a) ± 1 (b) ± 2 (c) $\pm 1/\sqrt{2}$ (d) none
22. If $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = (lx + my + n)(l'x + m'y + n')$ and $\Delta_1 = \begin{vmatrix} l & l' & 0 \\ m & m' & 0 \\ n & n' & 0 \end{vmatrix}$ and $\Delta_2 = \begin{vmatrix} l' & l & 0 \\ m' & m & 0 \\ n' & n & 0 \end{vmatrix}$, then the product $\Delta_1 \Delta_2$ is
 (a) $\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0$ (b) $\begin{vmatrix} h & g & f \\ a & f & c \\ b & c & h \end{vmatrix} = 0$ (c) 1 (d) none

23. If $f(x) = \frac{1}{6} \begin{vmatrix} \sin x & \cos x & \sin x \\ \sin 2x & \cos 2x & \sin 4x \\ \cos 3x & \sin 3x & \cos 7x \end{vmatrix}$, then $f'\left(\frac{\pi}{2}\right)$ is equal to
 (a) 1 (b) -1 (c) $1/3$ (d) none of these
24. If $a, b > 0$ and $\Delta = \begin{vmatrix} x & a & a \\ b & x & a \\ b & b & x \end{vmatrix}$, then Δ has a minimum at
 (a) $x = a$ (b) $x = b$ (c) $x = \sqrt{ab}$ (d) $x = 0$
25. If $P = \begin{bmatrix} x & y & z \end{bmatrix}$, $Q = \begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix}$, $R = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ then $[ax^2 + by^2 + cz^2 + 2hxy + 2gzx + 2fyz]$ is
 (a) $(QP)R$ (b) $(PQ)R$ (c) $(PR)Q$ (d) none of these

DAY-7

- If A and B are square matrices of equal size, then which one is correct ?
 (a) $A + B = B + A$ (b) $A + B = A - B$ (c) $A - B = B - A$ (d) $AB = BA$
- If A is singular matrix, then $\text{adj } A$ is :
 (a) singular (b) non-singular (c) symmetric (d) not defined
- A and B are two non-zero square matrices such that $AB = 0$. Then :
 (a) both A and B are singular (b) either of them is singular
 (c) neither matrix is singular (d) none of these
- From the matrix equation $AB = AC$, we can conclude $B = C$ provided :
 (a) A is singular (b) A is non-singular (c) A is symmetric (d) A is square
- A and B be square matrices of same size. Then $|A - B|$ implies :
 (a) $A = 0$ or $B = 0$ (b) $|A| = 0$ and $|B| = 0$ (c) $|A| = 0$ or $|B| = 0$ (d) none of these
- A and B are two square matrices of same order and A' denotes the transpose of A then :
 (a) $(AB)' = B' A'$ (b) $(AB)' = A' B'$ (c) $AB = 0 \Rightarrow A = 0$ or $B = 0$ (d) none of these
- If A and B are 3×3 matrices and $|A| \neq 0$ then which of the following are true ?
 (a) $|AB| = 0 \Rightarrow |B| = 0$ (b) $|A^{-1}| = |A|^{-1}$ (c) $|A + A| = 2 |A|$ (d) none of these
- If $A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & -1 & 0 \\ -1 & 0 & 0 \end{bmatrix}$, then
 (a) A is null matrix (b) $A = -1$ (c) A^{-1} does not exist (d) $A^2 = I$
- If A is 3×4 matrix and B is a matrix such that $A' B$ and BA' both are defined, then B is of the type :
 (a) 3×4 (b) 3×3 (c) 4×4 (d) 4×3
- If A is a singular matrix then $A \text{ adj } (A)$:
 (a) is a scalar matrix (b) is a zero matrix
 (c) is an identity matrix (d) is an orthogonal matrix
- A and B are square matrices of size $n \times n$ such that $A^2 - B^2 = (A - B)(A + B)$, then which of the following is always true ?
 (a) $A = B$ (b) $AB = BA$ (c) A or B is zero matrix (d) none of these
- If A and B are any two square matrix of the same order, then which of the following is/are true ?
 (a) $(AB)' = A' B'$ (b) $|AB| = 0$ if $|A| = 0$ or $|B| = 0$
 (c) $|AB| = 0$ if $|A| = I$ or $|B| = I$ (d) none of these
- If A is a skew-symmetric matrix of odd order, then $|\text{adj } A|$ is equal to :
 (a) 0 (b) n (c) n^2 (d) none of these
- 'A' is any square matrix, then $\det (A - A^T)^T$ is equal to :
 (a) 0 (b) 1
 (c) can be 0 or a perfect square (d) cannot be determined



15. Consider the following statements:
 (1) A square matrix A is Hermitian if $A = A'$.
 (2) Let $A = [a_{ij}]$ be a skew-Hermitian matrix, then a_{ii} is purely imaginary.
 (3) All integral powers of a symmetric matrix are symmetric.
 Which of these is / are correct?
 (a) 1 and 2 (b) 2 and 3 (c) 3 and 1 (d) 1, 2 and 3
16. If $A = \begin{bmatrix} ab & b^2 \\ -a^2 & -ab \end{bmatrix}$, then A is :
 (a) idempotent matrix (b) involuntary matrix
 (c) nilpotent matrix (d) scalar matrix
17. If A is an invertible symmetric matrix, then A^{-1} is :
 (a) symmetric (b) skew symmetric (c) a diagonal matrix (d) none of these.
18. If $A = [a_{ij}]$ is a skew-symmetric matrix of order n, then $a_{ii} =$
 (a) 0 for some i (b) 0 for all $i = 1, 2, \dots, n$
 (c) 1 for some i (d) 1 for all $i = 1, 2, \dots, n$
19. If A is a square matrix, then $A - A'$ is
 (a) symmetric (b) skew-symmetric (c) scalar (d) diagonal
20. If $A = [a_{ij}]$ is a scalar matrix, then trace of A is
 (a) $\sum_i \sum_j a_{ij}$ (b) $\sum_i a_{ii}$ (c) $\sum_j a_{ij}$ (d) $\sum_i a_{ii}$
21. A matrix is
 (a) a collection of real numbers (b) a collection of real or complex numbers
 (c) an array of real numbers (d) an array of real or complex numbers
22. If A is a skew-symmetric matrix and n is a positive integer, then A^n is
 (a) a symmetric matrix (b) skew-symmetric matrix
 (c) diagonal matrix (d) none of these
23. If A and B are invertible matrices, which of the following statement is not correct
 (a) $\text{Adj. } A = |A| A^{-1}$ (b) $\det(A^{-1}) = [\det(A)]^{-1}$
 (c) $(A+B)^{-1} = A^{-1} + B^{-1}$ (d) $(AB)^{-1} = B^{-1} A^{-1}$
24. If $A = \begin{bmatrix} \cos \alpha & \sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix}$ then A^2 is given by
 (a) $\begin{bmatrix} 1 & \cos 2\alpha \\ \sin 2\alpha & 0 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & \sin 2\alpha \\ \sin 2\alpha & 1 \end{bmatrix}$ (c) $\begin{bmatrix} 1 & \cos 2\alpha \\ \cos 2\alpha & 1 \end{bmatrix}$ (d) none of these
25. If $A = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$, then A^5 is
 (a) 4A (b) 5A (c) 8A (d) 16A

DAY-8

1. Rank of the matrix $\begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{bmatrix}$ equals
 (a) 0 (b) 1 (c) 2 (d) 3
2. Rank of the matrix $\begin{bmatrix} 1 & -3 & 2 \\ -3 & 9 & -6 \\ 2 & -6 & 4 \end{bmatrix}$ is
 (a) 0 (b) 1 (c) 2 (d) 3

3. If x, y, z are in A.P. with common difference d and the rank of the matrix $\begin{bmatrix} 4 & 5 & x \\ 5 & 6 & y \\ 6 & k & z \end{bmatrix}$ is 2, then the value of d and k are :
 (a) $x/4, k \in R$ (b) $d \in R, 7$ (c) $x, 5$ (d) $x/2, 6$
4. The rank of the matrix $\begin{bmatrix} -1 & 2 & 5 \\ 2 & -4 & a-4 \\ 1 & -2 & a+1 \end{bmatrix}$ is :
 (a) 1 if $a = -6$ (b) 2 if $a \neq -6$ (c) 3 if $a = 2$ (d) (a) or (b)
5. The rank of matrix $A = \frac{1}{3} \begin{bmatrix} 2 & 3 & 1 & 4 \\ 0 & 1 & 2 & -1 \\ 0 & -2 & -4 & 2 \end{bmatrix}$ is :
 (a) 2 (b) 3 (c) 1 (d) indeterminate
6. The rank of the matrix $\begin{bmatrix} 4 & 2 & 1-x \\ 5 & k & 1 \\ 6 & 3 & 1+x \end{bmatrix}$ is 2, then :
 (a) $k = \frac{5}{2}, x = \frac{1}{5}$ (b) $k = \frac{5}{2}, x \neq \frac{1}{5}$ (c) $k = \frac{1}{5}, x = \frac{5}{2}$ (d) none of these
7. If A and B are two matrices such that rank of A = m and rank of B = n , then :
 (a) rank (AB) = mn (b) rank (AB) $\geq$ rank (A)
 (c) rank (AB) $\geq$ rank (B) (d) rank (AB) $\leq$ minimum [rank (A), rank (B)]
8. The rank of the matrix $\begin{bmatrix} -5 & 3 & 2 \\ 3 & 2 & -5 \\ 4 & -1 & -3 \end{bmatrix}$ is
 (a) 3 (b) 2 (c) 1 (d) none of these
9. The rank of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ \lambda & 2 & 4 \\ 2 & -3 & 1 \end{bmatrix}$ is 3 if
 (a) $\lambda \neq \frac{18}{11}$ (b) $\lambda = \frac{18}{11}$ (c) $\lambda = -\frac{18}{11}$ (d) none of these
10. The rank of the matrix $\begin{bmatrix} 4 & 1 & 0 & 0 \\ 3 & 0 & 1 & 0 \\ 5 & 0 & 0 & 1 \end{bmatrix}$ is
 (a) 4 (b) 3 (c) 2 (d) none of these

DAY-9

1. If the equations $x + 2y + 3 = 0$, $2x + 3y + 5 = 0$ and $ax + by + 8 = 0$ are consistent, then $a + b$ is equal to:
 (a) 8 (b) 0 (c) 10 (d) none of these.
2. The value of a for which the system of equations $(a+1)^3 x + (a+2)^3 y = (a+3)^3$, $(a+1)x + (a+2)y = a+3$, $x + y = 1$ is consistent, is :
 (a) -2 (b) -1 (c) 1 (d) 2
3. If the system of equations $x - ky - z = 0$, $kx - y - z = 0$, $x + y - z = 0$ has a non-zero solution, then the possible values of k are :
 (a) -1, 2 (b) 1, 2 (c) 0, 1 (d) -1, 1
4. The equation $2x + y = 5$, $x + 3y = 5$, $x - 2y = 0$ have :
 (a) no solution (b) one solution (c) two solutions (d) infinite solutions

5. If the system of equations $x + 2y - 3z = 1$, $(p + 2)z = 3$, $(2p + 1)y + z = 2$ are inconsistent, then the value of p is :
 (a) -2 (b) 0 (c) 2 (d) None
6. The system of linear equations $x + y + z = 2$, $2x + y - z = 3$, $3x + 2y + kz = 4$ has a unique solution if :
 (a) $k \neq 0$ (b) $-1 < k < 1$ (c) $-2 < k < 2$ (d) $k = 0$
7. Consider the system of equations $a_1x + b_1y + c_1z = 0$, $a_2x + b_2y + c_2z = 0$, $a_3x + b_3y + c_3z = 0$ if $\begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix} = 0$, then the system has :
 (a) more than two solutions (b) one trivial and one non-trivial solutions
 (c) no solution (d) only trivial solution $(0, 0, 0)$
8. If the system of equations $x + 2y - 3z = 2(k + 3)z = 3$, $(2k + 1)y + z = 2$ is inconsistent, then k is :
 (a) -3 (b) $1/2$ (c) 1 (d) 2
9. The system of equations $\alpha x + y + z = \alpha - 1$, $x + \alpha y + z = \alpha - 1$, $x + y + \alpha z = \alpha - 1$ has no solution, if α is :
 (a) -2 (b) either -2 or 1 (c) not -2 (d) 1
10. If the system of equation $x + ay = 0$, $az + y = 0$ and $ax + z = 0$ has infinite solution, then the value of a is :
 (a) -1 (b) 1 (c) 0 (d) no real values
11. Given $2x - y + 2z = 2$, $x - 2y + z = -4$, $x + y + \lambda z = 4$, then the value of λ is :
 (a) 3 (b) 1 (c) 0 (d) -3
12. The value of a for which the system of equations $a^3x + (a + 1)^3y + (a + 2)^3z = 0$; $ax + (a + 1)y + (a + 2)z = 0$; $x + y + z = 0$ has a non-zero solution, is :
 (a) 1 (b) 0 (c) -1 (d) none
13. The system of equations $x + y + z = 6$, $x + 2y + 3z = 10$, $x + 2y + \lambda z = 12$ is inconsistent, if :
 (a) $\lambda = 1$ (b) $\lambda = 2$ (c) $\lambda = -2$ (d) $\lambda = 3$
14. For what value of k the following system of linear equations will have infinite solutions ?
 $x - y + z = 3$, $2x + y - z = 2$, $-3x - 2ky + 6z = 3$,
 (a) $k \neq 0$ (b) $k = 0$ (c) $k = 3$ (d) k
15. The number of values of k for which the system of equations $(k + 1)x + 8y = 4k$, $kx + (k + 3)y = 3k - 1$ has infinite many solution is :
 (a) 0 (b) 1 (c) 2 (d) infinite
16. If the system of equations $2x + 3y = 3$, $(c + 2)x + (c + 4)y = c + 6$, $(c + 2)^2x + (c + 4)^2y = (c + 6)^2$ is consistent, then $c =$
 (a) 0 (b) 10 (c) -10 (d) (a) or (c)
17. Consider the system of linear equations $a_1x + b_1y + c_1z + d_1 = 0$, $a_2x + b_2y + c_2z + d_2 = 0$ and $a_3x + b_3y + c_3z + d_3 = 0$, let us denote by $\Delta(a, b, c)$ as the determinant $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$, if $\Delta(a, b, c) \neq 0$, then the value of x in the unique solution of the above equations is
 (a) $\frac{\Delta(b, c, d)}{\Delta(a, b, c)}$ (b) $-\frac{\Delta(b, c, d)}{\Delta(a, b, c)}$ (c) $\frac{\Delta(a, c, d)}{\Delta(a, b, c)}$ (d) $-\frac{\Delta(a, b, d)}{\Delta(a, b, c)}$
18. For the system of equations $a_1x + b_1y + c_1z = 0$, $a_2x + b_2y + c_2z = 0$, and $a_3x + b_3y + c_3z = 0$.
 If $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0$, then the system has :
 (a) more than two solutions (b) one trivial and one non-trivial
 (c) no solution (d) only trivial solution $(0, 0, 0)$

19. The set of all values of λ for which the system of linear equations $\lambda x + (\sin \alpha)y + (\cos \alpha)z = 0$, $x + (\cos \alpha)y + (\sin \alpha)z = 0$, $-x + (\sin \alpha)y - (\cos \alpha)z = 0$ has a non-trivial solution is given by :
 (a) $[-2, 2]$ (b) $[-\sqrt{2}, \sqrt{2}]$ (c) $(-\sqrt{2}, \sqrt{2})$ (d) none of these
20. Real values of r for which the system of equations has non-trivial solutions $2rx - 2y + 3z = 0$, $x + ry + 2z = 0$, $2x + rz = 0$ are given by :
 (a) $2 - 1$ (b) -1 (c) 2 (d) none

LEVEL-2 (Single Correct Option)

1. If neither α nor β are multiple of $\pi/2$ and the product AB of matrices $A = \begin{bmatrix} \cos^2 \alpha & \cos \alpha \sin \alpha \\ \cos \alpha \sin \alpha & \sin^2 \alpha \end{bmatrix}$ and $B = \begin{bmatrix} \cos^2 \beta & \cos \beta \sin \beta \\ \cos \beta \sin \beta & \sin^2 \beta \end{bmatrix}$ is null matrix, then $\alpha - \beta$ is :
 (a) 0 (b) multiple of π (c) an odd multiple of $\pi/2$ (d) none of the above
2. If $(x_1 - x_2)^2 + (y_1 - y_2)^2 = a^2$, $(x_2 - x_3)^2 + (y_2 - y_3)^2 = b^2$ and $(x_3 - x_1)^2 + (y_3 - y_1)^2 = c^2$, then $4 \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}^2$ is equal to:
 (a) $abc(a+b+c)$ (b) $(a+b+c)^4$
 (c) $(a+b+c)(a+b-c)(b+c-a)(c+a-b)$ (d) none of the above.
3. If α, β, γ are the roots of $x^3 + ax^2 + b = 0$, then the value of $\begin{vmatrix} \alpha & \beta & \gamma \\ \beta & \gamma & \alpha \\ \gamma & \alpha & \beta \end{vmatrix}$ is :
 (a) $-a^3$ (b) $a^3 - 3b$ (c) a^3 (d) $a^2 - 3b$
4. If the lines $x + ay = a$, $bx + y = b$ and $cx + cy = 1$ are concurrent, then the value of $\frac{a}{1-a} + \frac{b}{1-b} + \frac{c}{1-c}$ is:
 (a) 0 (b) 1 (c) -1 (d) none of these.
5. If $\vec{r}_i = (a_i, b_i, c_i)$ where $i = 1, 2, 3$ be three mutually perpendicular unit vectors, then the value of $\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}^2$ is
 (a) 1 (b) 0 (c) 4 (d) none of these.
6. If $\Delta_r = \begin{vmatrix} 2r-1 & {}^nC_r & 1 \\ n^2 & 2^n-1 & n \\ a & b & c \end{vmatrix}$, then $\sum_{r=1}^n \Delta_r$ depends upon :
 (a) n only (b) a, b, c (c) a, c (d) none of these
7. If $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = -(a+b+c)(a+bk+ck^2)(a+bk^2+ck)$, then k equals to :
 (a) 1 (b) -1 (c) ω^2 (d) none of these.
8. The value of the determinant $\begin{vmatrix} \log_a \left(\frac{x}{y} \right) & \log_a \left(\frac{y}{z} \right) & \log_a \left(\frac{z}{x} \right) \\ \log_b \left(\frac{y}{z} \right) & \log_b \left(\frac{z}{x} \right) & \log_b \left(\frac{x}{y} \right) \\ \log_c \left(\frac{z}{x} \right) & \log_c \left(\frac{x}{y} \right) & \log_c \left(\frac{y}{z} \right) \end{vmatrix}$ is :
 (a) 1 (b) -1 (c) $\log_a xyz$ (d) none of these

9. The equations $ax + a^2y + z = 0$; $bx + b^2y + z = 0$; $cx + c^2y + z = 0$ have only zero solution if common ratio of G.P. is not equal to (here a, b, c are in G.P.)
 (a) $0, \pm 1$ (b) $2, \pm 3$ (c) $4, \pm 5$ (d) none of these
10. If a, b and c are the sides of a triangle and A, B, C are the angles opposite to a, b, c respectively then

$$\begin{vmatrix} a^2 & b \sin A & c \sin A \\ b \sin A & 1 & \cos A \\ c \sin A & \cos A & 1 \end{vmatrix}$$
 is
 (a) 0 (b) $\sum \frac{c \sin(A+B)}{a^2 + b^2}$ (c) 1 (d) none of these
11. If $i = \sqrt{-1}$ and $\sqrt[4]{1} = \alpha, \beta, \gamma, \delta$ then

$$\begin{vmatrix} \alpha & \beta & \gamma & \delta \\ \beta & \gamma & \delta & \alpha \\ \gamma & \delta & \alpha & \beta \\ \delta & \alpha & \beta & \gamma \end{vmatrix}$$
 is equal to
 (a) i (b) $-i$ (c) 1 (d) 0
12. If $a^{-1} + b^{-1} + c^{-1} = 0$ and if $D = \begin{vmatrix} 1+a & 1 & 1 \\ 1 & b+1 & 1 \\ 1 & 1 & 1+c \end{vmatrix}$, then D is
 (a) $abc \left(1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right)$ (b) abc (c) $a^2b^2c^2$ (d) none
13. The value of determinant

$$\begin{vmatrix} 1+a_1 & a_2 & a_3 & a_4 \\ a_1 & 1+a_2 & a_3 & a_4 \\ a_1 & a_2 & 1+a_3 & a_4 \\ a_1 & a_2 & a_3 & 1+a_4 \end{vmatrix}$$

 (a) $\prod a_i$ (b) $\frac{\sum a_i}{\prod a_i}$ (c) 0 (d) none of these
14. If $z = \begin{vmatrix} 3+3i & 5-i & 7-3i \\ i & 2i & 3i \\ 3-2i & 5+i & 7+3i \end{vmatrix}$, then
 (a) z is purely real (b) z is purely imaginary
 (c) z is mixed complex number with imaginary part positive (d) none of these
15. If α, β are non-real numbers satisfying $x^3 - 1 = 0$ then the value of

$$\begin{vmatrix} \lambda+1 & \alpha & \beta \\ \alpha & \lambda+\beta & 1 \\ \beta & 1 & \lambda+\alpha \end{vmatrix}$$
 is equal to
 (a) 0 (b) λ^3 (c) $\lambda^3 + 1$ (d) none of these
16. The determinant $\Delta = \begin{vmatrix} a^2+x & ab & ac \\ ab & b^2+x & bc \\ ac & bc & c^2+x \end{vmatrix}$ is divisible by
 (a) x (b) x^2 (c) $a^2 + b^2 + c^2 + x$ (d) none
17. If a, b, c, d are the roots of $x^4 - px^3 + qx^2 + rx + s = 0$, then $\Delta = \begin{vmatrix} 1+a & 1 & 1 & 1 \\ 1 & 1+b & 1 & 1 \\ 1 & 1 & 1+c & 1 \\ 1 & 1 & 1 & 1+d \end{vmatrix}$ is
 (a) $s-r$ (b) s/r (c) $s+r$ (d) none of these

18. If $\begin{vmatrix} b+c & c+a & a+b \\ a+b & b+c & c+a \\ c+a & a+b & b+c \end{vmatrix} = k \begin{vmatrix} a & b & c \\ c & a & b \\ b & c & a \end{vmatrix}$, then the value of k is
 (a) 1 (b) 2 (c) 3 (d) 4
19. If $f(x) = \Delta = \begin{vmatrix} \log x^2 & \cos x & x \\ x^4 & \sin x^2 & \sin x \\ 1 & e^{x^2} & x^3 \end{vmatrix}$, then the value of $\int_{-\pi/2}^{\pi/2} f(x) dx$ is
 (a) 0 (b) 1 (c) 2 (d) none
20. The value of determinant $\begin{vmatrix} \sqrt{3} + \sqrt{13} & 2\sqrt{5} & \sqrt{5} \\ \sqrt{15} + \sqrt{26} & 5 & \sqrt{10} \\ 3 + \sqrt{65} & \sqrt{15} & 5 \end{vmatrix}$ is
 (a) $15\sqrt{2} - 25\sqrt{3}$ (b) $15\sqrt{3} - 25\sqrt{2}$ (c) 0 (d) none of these
21. If $a > 0, b > 0$ and $c > 0$ are respectively p th, q th and r th terms of a G.P. then the value of determinant $\begin{vmatrix} \log a^2 & (p-1) & 3 \\ \log b^4 & 2(q-1) & 6 \\ \log c^8 & 4(r-1) & 12 \end{vmatrix}$, is
 (a) 1 (b) 0 (c) -1 (d) none of these
22. If the system of equation $2x + 3y + 4 = 0, 3x + 5y + 6 = 0$ and $2x^2 + 5y^2 + 6xy + 8x + 12y + 1 = t$ consistent of a solution, then value of t is
 (a) 4 (b) -4 (c) 1 (d) -1
23. The equations $ax + a^2y + z = 0, bx + b^2y + z = 0$ and $cx + c^2y + z = 0$ have only zero solution, then common ratio of G.P. is not equal to $\{a, b, c \text{ are in G.P.}\}$
 (a) 0, 1, -1 (b) 2, 3, -3 (c) 4, 5, -5 (d) none of these
24. When $\begin{vmatrix} y+z & z & y \\ z & z+x & x \\ y & x & x+y \end{vmatrix}$ is divided by xyz , then quotient will be :
 (a) 1 (b) 2 (c) 4 (d) none of these
25. Given that $\begin{vmatrix} b^2 + c^2 & ab & ca \\ ab & c^2 + a^2 & bc \\ ca & bc & a^2 + b^2 \end{vmatrix} = 4k$, then k is equal to :
 (a) $a^2b^2c^2$ (b) ab^2c^3 (c) bc^2a^3 (d) ca^2b^3
26. For all values of A, B, C and P, Q, R, $\begin{vmatrix} \cos(A-P) & \cos(A-Q) & \cos(A-R) \\ \cos(B-P) & \cos(B-Q) & \cos(B-R) \\ \cos(C-P) & \cos(C-Q) & \cos(C-R) \end{vmatrix} =$
 (a) 0 (b) $\cos A \cos B \cos C$ (c) 1 (d) $\sin A \sin B \sin C$
27. For a fixed positive integer n , if $\Delta = \begin{vmatrix} n! & (n+1)! & (n+2)! \\ (n+1)! & (n+2)! & (n+3)! \\ (n+2)! & (n+3)! & (n+4)! \end{vmatrix}$; then $\left[\frac{\Delta}{(n!)^3} - 4 \right]$ is divisible by:
 (a) n (b) n^2 (c) $n(n+1)$ (d) none
28. Suppose three digit numbers A28, 3 B9, 62C; A, B, C are integers between 0 and 9, are divisible by a fixed integer k . Then the determinant $\begin{vmatrix} A & 3 & 6 \\ 8 & 9 & C \\ 2 & B & 2 \end{vmatrix}$ is divisible by :
 (a) k^2 (b) $2k$ (c) k (d) none

29.
$$\begin{vmatrix} \sin\theta & \cos\theta & \sin 2\theta \\ \sin\left(\theta + \frac{2\pi}{3}\right) & \cos\left(\theta + \frac{2\pi}{3}\right) & \sin\left(2\theta + \frac{4\pi}{3}\right) \\ \sin\left(\theta - \frac{2\pi}{3}\right) & \cos\left(\theta - \frac{2\pi}{3}\right) & \sin\left(2\theta - \frac{4\pi}{3}\right) \end{vmatrix} =$$
- (a) $3\sin\theta$ (b) $\sin^3\theta$ (c) 0 (d) none
30. If $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 5$, then the value of $\begin{vmatrix} b_2c_3 - b_3c_2 & a_3c_2 - a_2c_3 & a_2b_3 - a_3b_2 \\ b_3c_1 - b_1c_3 & a_1c_3 - a_3c_1 & a_3b_1 - a_1b_3 \\ b_1c_2 - b_2c_1 & a_2c_1 - a_1c_2 & a_1b_2 - a_2b_1 \end{vmatrix}$ is :
- (a) 5 (b) 25 (c) 125 (d) 0
31. If $a_1, a_2, a_3, \dots, a_n, \dots$ are in G.P. then the value of the determinant $\begin{vmatrix} \log a_n & \log a_{n+1} & \log a_{n+2} \\ \log a_{n+3} & \log a_{n+4} & \log a_{n+5} \\ \log a_{n+6} & \log a_{n+7} & \log a_{n+8} \end{vmatrix}$, is
- (a) 0 (b) 1 (c) 2 (d) -2
32. If $\Delta_r = \begin{vmatrix} x & y & z \\ 2^r & 2 \times 3^r & 3 \times 4^r \\ 2 \times (2^n - 1) & 3 \times (3^n - 1) & 4 \times (4^n - 1) \end{vmatrix}$, then $\sum_{r=1}^n \Delta_r$ is equal to :
- (a) 0 (b) xyz (c) $24xyz(2^n - 1)(3^n - 1)(4^n - 1)$ (d) none of these
33. The value of the determinant $\begin{vmatrix} \sin^2\left(x + \frac{3\pi}{2}\right) & \sin^2\left(x + \frac{5\pi}{2}\right) & \sin^2\left(x + \frac{7\pi}{2}\right) \\ \sin\left(x + \frac{3\pi}{2}\right) & \sin\left(x + \frac{5\pi}{2}\right) & \sin\left(x + \frac{7\pi}{2}\right) \\ \sin\left(x - \frac{3\pi}{2}\right) & \sin\left(x - \frac{5\pi}{2}\right) & \sin\left(x - \frac{7\pi}{2}\right) \end{vmatrix}$ is
- (a) 1 (b) 2 (c) 3 (d) 0
34. If α, β, γ are the roots of $px^3 + qx^2 + r = 0$ then the value of the determinant $\begin{vmatrix} \alpha\beta & \beta\gamma & \gamma\alpha \\ \beta\gamma & \gamma\alpha & \alpha\beta \\ \gamma\alpha & \alpha\beta & \beta\gamma \end{vmatrix}$ is
- (a) pq (b) qr (c) 0 (d) pr
35. If $[x]$ stands for the greatest integer less than or equal to x , then the value of $\begin{vmatrix} [e] & [\pi] & [\pi^2 - 6] \\ [\pi] & [\pi^2 - 6] & [e] \\ [\pi^2 - 6] & [e] & [\pi] \end{vmatrix}$ is
- (a) -8 (b) 8 (c) 0 (d) 1
36. The determinant $\begin{vmatrix} \cos C & \tan A & 0 \\ \sin B & 0 & -\tan A \\ 0 & \sin B & \cos C \end{vmatrix}$ has the value, where A, B, C are angles of a triangle,
- (a) 0 (b) 1 (c) $\sin A \sin B$ (d) $\cos A \cos B \cos C$
37. If $f = \begin{vmatrix} x & y & z \\ x^2 & y^2 & z^2 \\ yz & zx & zy \end{vmatrix}$ and $g = (x-y)(y-z)(z-x)$, then $\frac{f}{g}$ is :
- (a) $xy + yz + zx$ (b) $x^2 + y^2 + z^2$ (c) $x^2 + y^2 + z^2 - xy - yz - zx$ (d) none of these

38. When the determinant $\begin{vmatrix} \cos 2x & \sin^2 x & \cos 4x \\ \sin^2 x & \cos 2x & \cos^2 x \\ \cos 4x & \cos^2 x & \cos 2x \end{vmatrix}$ is expanded in powers of $\sin x$, then the constant term in that expansion is
 (a) 1 (b) 0 (c) -1 (d) 2
39. The value of the determinant $\begin{vmatrix} \sin \theta & \cos \theta & \sin 2\theta \\ \sin\left(\theta + \frac{2\pi}{3}\right) & \cos\left(\theta + \frac{2\pi}{3}\right) & \sin\left(2\theta + \frac{4\pi}{3}\right) \\ \sin\left(\theta - \frac{2\pi}{3}\right) & \cos\left(\theta - \frac{2\pi}{3}\right) & \sin\left(2\theta - \frac{4\pi}{3}\right) \end{vmatrix}$ is
 (a) 0 (b) $2 \sin \theta$ (c) $\sin 2\theta$ (d) none of these
40. If $g(x) = \begin{vmatrix} e^{-x} & e^{x \log_e a} & x^2 \\ a^{-3x} & e^{3x \log_e a} & x^4 \\ a^{-5x} & e^{5x \log_e a} & 1 \end{vmatrix}$, then
 (a) $g(x) + g(-x) = 0$ (b) $g(x) - g(-x) = 0$ (c) $g(x) \times g(-x) = 0$ (d) none of these
41. If $0 \leq [x] < 2$; $-1 \leq [y] < 1$ and $1 \leq [z] < 3$ ($[.]$ denotes the greatest integer function) then the maximum value of determinant $\Delta = \begin{vmatrix} [x]+1 & [y] & [z] \\ [x] & [y]+1 & [z] \\ [x] & [y] & [z]+1 \end{vmatrix}$ is
 (a) 2 (b) 6 (c) 4 (d) none of these
42. A determinant of second order is made with the elements 0 and 1. The number of determinants with non-negative values is
 (a) 3 (b) 10 (c) 11 (d) 13
43. Let $\{D_1, D_2, D_3, \dots, D_n\}$ be the set of third order determinants that can be made with the distinct non-zero real numbers $a_1, a_2, \dots, a_n$ (using all), then
 (a) $\sum_{i=1}^n D_i = 1$ (b) $\sum_{i=1}^n D_i = 0$ (c) $D_i = D_j, \forall i, j$ (d) none of these
44. If $x_i = a_i b_i c_i, i = 1, 2, 3$ are three-digit positive integer such that each x_i is a multiple of 19, then for some integer n
 $\Delta = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$ is given by
 (a) $19n+1$ (b) $19n+2$ (c) $19n$ (d) $19n+3$
45. If a, b, c , are the sides of a triangle ABC such that $\begin{vmatrix} a^2 & b^2 & c^2 \\ (a+1)^2 & (b+1)^2 & (c+1)^2 \\ (a-1)^2 & (b-1)^2 & (c-1)^2 \end{vmatrix} = 0$, then
 (a) ΔABC is non-isosceles right-angled triangle
 (b) ΔABC is an equilateral
 (c) ΔABC is an acute angled triangle with non two angles being equal
 (d) ΔABC is an isosceles
46. Let $\Delta = \begin{vmatrix} \alpha & \alpha+\beta & \alpha+\beta+\gamma \\ 3\alpha & 4\alpha+3\beta & 5\alpha+4\beta+3\gamma \\ 6\alpha & 9\alpha+6\beta & 11\alpha+9\beta+6\gamma \end{vmatrix}$
 If α, β, γ represent three cube roots of unity then the value of Δ is
 (a) 0 (b) -1 (c) 1 (d) ω^2

47. If $\Delta = \begin{vmatrix} 1 & 3\cos\theta & 1 \\ \sin\theta & 1 & 3\cos\theta \\ 1 & \sin\theta & 1 \end{vmatrix}$, then maximum value of Δ is
 (a) 1 (b) 9 (c) 16 (d) 10
48. Coefficient of x in $f(x) = \begin{vmatrix} x & (1+\sin x)^3 & \cos x \\ 1 & \log(1+x) & 2 \\ x^2 & (1+x^2) & 0 \end{vmatrix}$ is
 (a) 0 (b) 1 (c) -2 (d) can't be determined
49. If a, b, c are the sides of a triangle $A B C$ and A, B, C are angles opposite to a, b, c and
 $\Delta = \begin{vmatrix} a^2 & b\sin A & c\sin A \\ b\sin A & 1 & \cos A \\ c\sin A & \cos A & 1 \end{vmatrix}$ then,
 (a) Δ = area of triangle (b) Δ = perimeter of the triangle
 (c) $\Delta = \sum a^2$ (d) none of these
50. If the equations $a_1x^2 + b_1x + c_1 = 0$ and $a_2x^2 + b_2x + c_2 = 0$ possess a common root, then $\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} \begin{vmatrix} b_1 & c_1 \\ b_2 & c_2 \end{vmatrix}$ is
 equal to :
 (a) $\begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix}$ (b) $2 \cdot \begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix}$ (c) $\begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix}^2$ (d) none of these
51. If $[p]$ stands for the greatest integer less than or equal to p , then in order that the set of equations,
 $2x - 3y = 4$, $7x - 2y = 2$, $[3\pi]x - [2e]y = [4a]$ may be consistent is that a should lie in :
 (a) $\left[\frac{1}{2}, 1\right]$ (b) $[2, 3]$ (c) $\left[\frac{3}{2}, \frac{7}{4}\right]$ (d) $\left[1, \frac{3}{2}\right]$
52. If t is real and $\lambda = \frac{t^2 - 3t + 4}{t^2 + 3t + 4}$, then the number of solutions of the system of equations
 $3x - y + 4z = 3$, $x + 2y - 3z = -2$, $6x + 5y + \lambda z = -3$ is :
 (a) one (b) two (c) zero (d) infinite
53. The set of equations : $\lambda x - y + (\cos\theta)z = 0$; $3x + 2y + 2z = 0$; $(\cos\theta)x + y + 2z = 0$
 $0 \leq \theta \leq 2\pi$, has non-trivial solutions
 (a) for no value of λ and θ (b) for all values of λ and θ
 (c) for all values of λ and only two values of θ (d) for only one value of λ and all values of θ
54. The set of homogeneous equations : $tx + (t+1)y + (t-1)z = 0$; $(t+1)x + ty + (t+2)z = 0$; $(t-1)x + (t+2)y + tz = 0$
 has non-trivial solution for
 (a) three values of t (b) two values of t (c) one value of t (d) no value of t
55. If a, b, c are non-zeros, then the system of equations : $(\alpha+a)x + \alpha y + \alpha z = 0$; $\alpha x + (\alpha+b)y + \alpha z = 0$;
 $\alpha x + \alpha y + (\alpha+c)z = 0$ has a non-trivial solution if :
 (a) $\alpha^{-1} = -(\alpha^{-1} + b^{-1} + c^{-1})$ (b) $\alpha^{-1} = a + b + c$
 (c) $\alpha + a + b + c = 1$ (d) none of the above
56. The set of linear equations : $x + 2\omega y + 3\omega^2 z = (1+\omega)(1-\omega)$; $2x + 3\omega y + \omega^2 z = \omega(\omega-1)$; $3x + \omega y + 2\omega^2 z = \omega-1$
 Where $\omega \neq 1$ is a cube root of unity, has the unique solution
 (a) $(1, 1, 1)$ (b) $(1, -1, 1)$ (c) $(1, -1, -1)$ (d) $(-1, -1, -1)$
57. If the system of equations $-ax + y + z = 0$, $x - by + z = 0$ and $x + y - cz = 0$ ($a, b, c \neq -1$) has a non-zero solution,
 then $\frac{1}{1+a} + \frac{1}{1+b} + \frac{1}{1+c} =$
 (a) 0 (b) 1 (c) 2 (d) 3

58. The value of x , so that $\begin{bmatrix} 1 & 3 & 2 \\ 0 & 5 & 1 \\ 0 & 3 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ x \end{bmatrix} = 0$ is
- (a) $\frac{-7 \pm \sqrt{35}}{2}$ (b) $\frac{-9 \pm \sqrt{53}}{2}$ (c) ± 2 (d) 0
59. The matrix X for which $\begin{bmatrix} 1 & -4 \\ 3 & -2 \end{bmatrix} X = \begin{bmatrix} -16 & -6 \\ 7 & 2 \end{bmatrix}$; is :
- (a) $\begin{bmatrix} -2 & 4 \\ -3 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} -\frac{1}{5} & \frac{2}{5} \\ -\frac{3}{10} & \frac{1}{5} \end{bmatrix}$ (c) $\begin{bmatrix} 6 & 2 \\ \frac{11}{2} & 2 \end{bmatrix}$ (d) $\begin{bmatrix} -16 & -6 \\ 7 & 2 \end{bmatrix}$
60. The value of a, b, c when $\begin{bmatrix} 0 & 2b & c \\ a & b & -c \\ a & -b & c \end{bmatrix}$ is orthogonal, are
- (a) $\pm \frac{1}{\sqrt{2}}, \pm \frac{1}{\sqrt{6}}, \pm \frac{1}{\sqrt{2}}$ (b) $\pm \frac{1}{\sqrt{3}}, \pm \frac{1}{\sqrt{2}}, \pm \frac{1}{\sqrt{6}}$
(c) $\pm \frac{1}{\sqrt{2}}, \pm \frac{1}{\sqrt{6}}, \pm \frac{1}{\sqrt{3}}$ (d) $\pm \frac{1}{\sqrt{2}}, \pm \frac{1}{\sqrt{2}}, \pm \frac{1}{\sqrt{2}}$
61. If $A = \begin{bmatrix} 1 & \tan x \\ -\tan x & 1 \end{bmatrix}$ then $A^T A^{-1} =$
- (a) $\begin{bmatrix} \cos 2x & -\sin 2x \\ \sin 2x & \cos 2x \end{bmatrix}$ (b) $\begin{bmatrix} -\cos 2x & \sin 2x \\ -\sin 2x & \cos 2x \end{bmatrix}$
(c) $\begin{bmatrix} \sin 2x & \cos 2x \\ \cos 2x & \sin 2x \end{bmatrix}$ (d) none of these
62. If $\begin{bmatrix} 2 & -1 \\ 1 & 0 \\ -3 & 4 \end{bmatrix} A = \begin{bmatrix} -1 & -8 & -10 \\ 1 & -2 & -5 \\ 9 & 22 & 15 \end{bmatrix}$, Then $A =$
- (a) $\begin{bmatrix} 2 & 1 & -3 \\ -1 & 0 & 4 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & -2 & -5 \\ 3 & 4 & 0 \end{bmatrix}$ (c) $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ (d) none of these
63. If $A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ p & q & r \end{bmatrix}$ then $A^3 - rA^2 - qA =$
- (a) pI (b) qI
(c) rI (d) none of these, I is third order unit matrix
64. X, Y and Z are positive numbers greater than 10 such that Y and Z have respectively 1 and 0 at their unit's place and Δ is the determinant $\begin{vmatrix} X & 4 & 1 \\ Y & 0 & 1 \\ Z & 1 & 0 \end{vmatrix}$. If $(\Delta + 1)$ is divisible by 10 then X has as its unit's place
- (a) 1 (b) 0 (c) 2 (d) none of these
65. For each real $x: -1 < x < 1$. Let $A(x)$ be the matrix $(1-x)^{-1} \begin{bmatrix} 1 & -x \\ -x & 1 \end{bmatrix}$ and $z = \frac{x+y}{1+xy}$ then
- (a) $A(z) = A(x) + A(y)$ (b) $A(z) = A(x) [A(y)]^{-1}$ (c) $A(z) = A(x) A(y)$ (d) $A(z) = A(x) - A(y)$

66. If P, Q and R represent the angles of an acute angled triangle, no two of them being equal then the value of

$$\begin{vmatrix} 1 & 1+\sin P & \sin P(1+\sin P) \\ 1 & 1+\sin Q & \sin Q(1+\sin Q) \\ 1 & 1+\sin R & \sin R(1+\sin R) \end{vmatrix}$$

- (a) 0 (b) positive (c) negative (d) can not be determined

67. $\begin{vmatrix} \operatorname{cosec} \alpha & 1 & 1 \\ 1 & 2 \operatorname{cosec} \alpha & 1 \\ 0 & 0 & 2 \operatorname{cosec} \alpha \end{vmatrix}$ gives the value equivalent to

- (a) $\frac{1}{2}(\tan^2 \alpha + \cot^2 \alpha)$ (b) $\frac{1}{2}(\tan^3 \alpha + \cot^3 \alpha)$
(c) $\frac{1}{2}\left(\tan^2 \frac{\alpha}{2} + \cot^2 \frac{\alpha}{2}\right)$ (d) $\frac{1}{2}\left(\tan^3 \frac{\alpha}{2} + \cot^3 \frac{\alpha}{2}\right)$

68. If α, β and γ are three different numbers $\begin{vmatrix} \alpha & \alpha^3 & \alpha^4 - 1 \\ \beta & \beta^3 & \beta^4 - 1 \\ \gamma & \gamma^3 & \gamma^4 - 1 \end{vmatrix} = 0$ is true iff

- (a) $\alpha + \beta + \gamma = \alpha\beta\gamma$ (b) $\alpha\beta + \beta\gamma + \gamma\alpha = \alpha\beta\gamma$
(c) $\alpha\beta\gamma(\alpha + \beta + \gamma) = \alpha\beta + \beta\gamma + \gamma\alpha$ (d) $\alpha\beta\gamma(\alpha\beta + \beta\gamma + \gamma\alpha) = \alpha + \beta + \gamma$

69. If $\begin{vmatrix} x & y & \alpha x - y \\ y & z & \alpha y - z \\ 2 & 1 & 0 \end{vmatrix} = 0$ and $\alpha \neq \frac{1}{2}$, then x, y, z are in

- (a) A.P. (b) G.P. (c) H.P. (d) none of these

70. The value of determinant $\begin{vmatrix} 1 & ka & k^2 + a^2 \\ 1 & kb & k^2 + b^2 \\ 1 & kc & k^2 + c^2 \end{vmatrix}$ is :

- (a) $k(a+b)(b+c)(c+a)$ (b) $k(a-b)(b-c)(c-a)$
(c) $kabc(a^2 + b^2 + c^2)$ (d) $k(a+b-c)(b+(c-a)(c+a-b))$

71. The system of equations

$$x + y + z = 2$$

$$2x - y + 3z = 5$$

$$x - 2y - z + 1 = 0$$

written in matrix form is

(a) $\begin{bmatrix} x \\ y \\ z \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 2 & -1 & 3 \\ 1 & -2 & -1 \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \\ -1 \end{bmatrix}$

(c) $\begin{bmatrix} 1 & 1 & 1 \\ 2 & -1 & 3 \\ 1 & -2 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \\ -1 \end{bmatrix}$

(b) $\begin{bmatrix} 1 & 1 & 1 \\ 2 & -1 & 3 \\ 1 & -2 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -2 \\ -5 \\ 1 \end{bmatrix}$

- (d) none of these

72. $\begin{vmatrix} \sec x & \cos x & \sec^2 x + \cot x \operatorname{cosec} x \\ \cos^2 x & \cos^2 x & \operatorname{cosec}^2 x \\ 1 & \cos^2 x & \cos^2 x \end{vmatrix}$ equals

- (a) $\sin^2 x + \cos^5 x$ (b) $-(\sin^2 x + \cos^5 x)$ (c) 0 (d) 1

73. If $\begin{bmatrix} 2 & 3 & 2 \\ 1 & x & 1 \\ 0 & 3 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ x \end{bmatrix} = 0$, then values of x are

- (a) 1, 8 (b) -1, 8 (c) -1, -8 (d) 1, -8

74. If three linear equations $x + 4ay + az = 0$, $x + 3by + bz = 0$ and $x + 2cy + cz = 0$ have a non-trivial solution, then a, b, c are in :
 (a) H.P. (b) G.P. (c) A.P. (d) none
75. If the system of linear equations $x + 2ay + az = 0$, $x + 3by + bz = 0$, $x + 4cy + cz = 0$ has a non-zero solution, then a, b, c :
 (a) are in A.P. (b) are in G.P. (c) are in H.P. (d) satisfy $a + 2b + 3c = 0$

LEVEL -3 (Multiple Correct Option(s))

1. The determinant $\begin{vmatrix} \cos(\theta + \phi) & -\sin(\theta + \phi) & \cos 2\phi \\ \sin \theta & \cos \theta & \sin \phi \\ -\cos \theta & \sin \theta & \cos \phi \end{vmatrix}$ is
 (a) non-negative (b) Independent of θ
 (c) Independent of ϕ (d) Independent of θ and ϕ both
2. If $\Delta = \begin{vmatrix} e^x & \sin x & 1 \\ \cos x & \log_e(1+x^2) & 1 \\ x & x^2 & 1 \end{vmatrix} = a + bx + cx^2$ then
 (a) $a = 0$ (b) $a = 1$ (c) $b = -1$ (d) $b = -2$
3. The value of x for which $\begin{vmatrix} x & 2 & 2 \\ 3 & x & 2 \\ 3 & 3 & x \end{vmatrix} + \begin{vmatrix} 1-x & 2 & 4 \\ 2 & 4-x & 8 \\ 4 & 8 & 16-x \end{vmatrix} > 33$ is
 (a) $0 < x < 1$ (b) $-\frac{1}{2} < x < \frac{1}{2}$ (c) $x < -\frac{1}{7}$ (d) $x > 1$
4. Let $f(n) = \begin{vmatrix} n & n+1 & n+2 \\ {}^nP_n & {}^{n+1}P_{n+1} & {}^{n+2}P_{n+2} \\ {}^nC_n & {}^{n+1}C_{n+1} & {}^{n+2}C_{n+2} \end{vmatrix}$ where the symbols have their usual meanings. Then $f(n)$ is divisible by
 (a) $n^2 + n + 1$ (b) $(n+1)!$ (c) $n!$ (d) none of these
5. If $f(x) = \begin{vmatrix} (1+x)^a & (1+2x)^b & 1 \\ 1 & (1+x)^a & (1+2x)^b \\ (1+2x)^b & 1 & (1+x)^a \end{vmatrix}$, a, b being positive integers, then
 (a) constant term of $f(x)$ is 0 (b) coefficient of x in $f(x)$ is 0
 (c) constant term in $f(x)$ is $a - b$ (d) constant term if $f(x)$ is $a + b$
6. If, a square matrix $A = [a_{ij}]$, $a_{ij} = i^2 - j^2$ is of even order then
 (a) A is a skew-symmetric (b) $|A|$ is a perfect square
 (c) A is symmetric and $|A| = 0$ (d) A is neither symmetric nor skew symmetric
7. If maximum and minimum values of the determinant $\begin{vmatrix} 1 + \sin^2 x & \cos^2 x & \sin 2x \\ \sin^2 x & 1 + \cos^2 x & \sin 2x \\ \sin^2 x & \cos^2 x & 1 + \sin 2x \end{vmatrix}$ are α and β , then
 (a) $\alpha + \beta^{99} = 4$
 (b) $\alpha^3 - \beta^{17} = 26$
 (c) $(\alpha^{2n} - \beta^{2n})$ is always an even integer for $n \in \mathbb{N}$
 (d) a triangle can be constructed having its sides as $\alpha - \beta$, $\alpha + \beta$ and $\alpha + 3\beta$
8. If $\begin{bmatrix} 1 & -\tan \theta \\ \tan \theta & 1 \end{bmatrix} \begin{bmatrix} 1 & \tan \theta \\ -\tan \theta & 1 \end{bmatrix}^{-1} = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$ then
 (a) $a = \cos 2\theta$ (b) $a = 1$ (c) $b = \sin 2\theta$ (d) $b = 1$

9. If $\begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 2 & 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 2 \\ 1 \end{bmatrix}$ then

(a) $x=1$ (b) $y=2$ (c) $z=3$ (d) none of these

10. If $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{40} \begin{bmatrix} 5 & 10 & -5 \\ -5 & -2 & 13 \\ 10 & -4 & 6 \end{bmatrix} \begin{bmatrix} 5 \\ 0 \\ 5 \end{bmatrix}$ then

(a) $x=0$ (b) $x+y+z=3$ (c) $y=0$ (d) none of these

11. If $\Delta_i = \begin{vmatrix} 2^{i-1} & \frac{1}{i(i+1)} & \sin i\theta \\ x & y & z \\ 2^n - 1 & \frac{n}{n+1} & \frac{\sin\left(\frac{n+1}{2}\theta\right) \sin \frac{n\theta}{2}}{\sin \frac{\theta}{2}} \end{vmatrix}$, then $\sum_{i=1}^n \Delta_i$ is

(a) 0 (b) independent of n
(c) independent of θ (d) independent of x, y and z

12. The determinant $\Delta = \begin{vmatrix} a^2 + x^2 & ab & ac \\ ab & b^2 + x^2 & bc \\ ac & bc & c^2 + x^2 \end{vmatrix}$ is divisible by

(a) x (b) x^2 (c) x^3 (d) x^4

13. The value of determinant $\begin{vmatrix} \sqrt{6} & 2i & 3+\sqrt{6} \\ \sqrt{12} & \sqrt{3}+\sqrt{8}i & 3\sqrt{2}+\sqrt{6}i \\ \sqrt{18} & \sqrt{2}+\sqrt{12}i & \sqrt{27}+2i \end{vmatrix}$ is a

(a) complex number (b) real number (c) irrational number (d) rational number

14. If $a > b > c$ and the system of equations $ax + by + cz = 0$, $bx + cy + az = 0$ and $cx + ay + bz = 0$ has a non trivial solution

(a) real (b) of opposite sign (c) positive (d) complex

15. Let a, b, c are non zero real and $\Delta = \begin{vmatrix} bc & ca & ab \\ ca & ab & bc \\ ab & bc & ca \end{vmatrix}$, Δ vanishes if

(a) $\frac{1}{a} + \frac{1}{b\omega} + \frac{1}{c\omega^2} = 0$

(b) $\frac{1}{a} + \frac{1}{b\omega^2} + \frac{1}{c\omega} = 0$

(c) $\frac{1}{a\omega} + \frac{1}{b\omega^2} + \frac{1}{c} = 0$

(d) $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 0$

16. The determinant $\begin{vmatrix} 2x-3 & x-2 & x-1 \\ x-2 & 2x-2 & x \\ x-1 & x & 2x-1 \end{vmatrix}$ vanishes if

(a) $x=0$

(b) $x = \frac{1}{2}$

(c) $x=1$

(d) $x=2$

17. $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ is

(a) a unit matrix

(b) identity matrix

(c) a diagonal matrix

(d) a scalar matrix

18. A matrix $A = [a_{ij}]_{m \times n}$ is
 (a) a horizontal matrix if $m > n$ (b) a horizontal matrix if $m < n$
 (c) a vertical matrix if $m > n$ (d) a vertical matrix if $m < n$
19. Choose the correct statement(s) about multiplication of two matrices A and B
 (a) whenever AB exists, it is not always necessary that BA also exist
 (b) whenever AB and BA both exist is always not necessary that they should be matrices of the same type (order)
 (c) when AB and BA both exist and are of the same order, then $AB = BA$
 (d) AB is never equal to BA
20. Which of the following is correct
 (a) If A is a square matrix $(A + A')$ is a symmetric matrix
 (b) If A is a square matrix $(A - A')$ is a skew symmetric matrix
 (c) Every square matrix can be expressed as the sum of a symmetric and a skew symmetric matrix
 (d) some elements of the skew symmetric matrix must be zero.
21. If $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{bmatrix}$, $B = \begin{bmatrix} -4 & -4 \\ -1 & 3 \end{bmatrix}$, $C = \begin{bmatrix} 3 & 2 & 1 \end{bmatrix}$, then which of the following exist
 (a) AB (b) BA (c) AC (d) CA
22. If $A = \begin{bmatrix} \cos \alpha & -\sin \alpha & 0 \\ \cos \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$ then
 (a) $A^{-1} = \text{Adj } A$ (b) $\text{Adj } A$ has four "zero" elements
 (c) A^{-1} does not exist if $\alpha = n\pi$ (d) $\text{Adj } A = A^T$
23. If A and B are invertible square of the same order, then which of the following is correct.
 (a) $\text{adj}(AB) = (\text{adj } A)(\text{adj } B)$ (b) $(\text{adj } A)' = (\text{adj } A')$
 (c) $|\text{adj } A| = |A|^{n-1}$ (d) $\text{adj}(\text{adj } B) = |B|^{n-2} B$.
24. If $A = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, $B = \begin{bmatrix} 4 & 5 & 6 \end{bmatrix}$, $C = \begin{bmatrix} 0 \\ -2 \\ 4 \end{bmatrix}$ then
 (a) $AB = \begin{bmatrix} 4 & 5 & 6 \\ 8 & 10 & 12 \\ 12 & 15 & 18 \end{bmatrix}$ (b) $AB = [32]$ (c) $(ABC)^T = \begin{bmatrix} 14 & 28 & 18 \end{bmatrix}$ (d) ABC is not defined
25. If $A = \begin{bmatrix} 3 & 1 \\ -1 & 2 \\ 0 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} 5 & 4 & 6 \\ 4 & 1 & 2 \\ -5 & -1 & 1 \end{bmatrix}$ then
 (a) $A + B$ exists (b) AB exists (c) BA exists (d) none of these
26. If $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ then
 (a) $A^3 = 9A$ (b) $A^3 = 27A$ (c) $A + A = A^2$ (d) A^{-1} does not exist
27. The digits A, B, C are such that the three digit numbers A88, 6B8, 86C are divisible by 72, then the determinant
 $\begin{vmatrix} A & 6 & 8 \\ 8 & B & 6 \\ 8 & 8 & C \end{vmatrix}$ is divisible by
 (a) 72 (b) 144 (c) 288 (d) 216

28. If $D_k = \begin{vmatrix} 1 & n & n \\ 2^{k-1} & 2^{n-1} & 2^{2n}-1 \\ 2 \cdot 3^{k-1} & 3^{n-1} & 3^{2n}-1 \end{vmatrix}$ then $\sum_{k=1}^n D_k$ is divisible by ($n > 1$)
- (a) 2^n (b) $2^n - 1$ (c) 3^n (d) $3^n - 1$
29. The determinant $\begin{vmatrix} a^3 & a^2 - bc & 1 \\ b^3 & b^2 - ca & 1 \\ c^3 & c^2 - ab & 1 \end{vmatrix}$ is divisible by
- (a) $(a-b)(b-c)(c-a)$ (b) $a^2 + b^2 + c^2 - ab - bc - ca$
(c) $a^2 + b^2 + c^2 + ab + bc + ca$ (d) $(a+b+c)^2$
30. The value of θ lying between $\theta = 0$ and $\theta = \pi/2$ and satisfying the equation $\begin{vmatrix} 1 + \sin^2 \theta & \cos^2 \theta & 4 \sin 4\theta \\ \sin^2 \theta & 1 + \cos^2 \theta & 4 \sin 4\theta \\ \sin^2 \theta & \cos^2 \theta & 1 + 4 \sin 4\theta \end{vmatrix} = 0$ are
- (a) $\frac{7\pi}{24}$ (b) $\frac{5\pi}{24}$ (c) $\frac{11\pi}{24}$ (d) $\frac{\pi}{24}$
31. The system of equations $ax + by + (a\lambda - b)z = 0$, $2ax + (c+a)y + (2a\lambda - c - a)z = 0$, $x + y = 0$ has non trivial solution if
- (a) a, b, c are in A.P. (b) a, b, c are in G.P. (c) $a = 0$ (d) $\lambda = 1$
32. If a homogenous system of 3 simultaneous equations in 3 unknowns is given then
- (a) Such a system is always consistent (b) It has infinity of solution
(c) Trivial solution always exists (d) Non trivial solutions always exist
33. The system of equations $2x + ky = 2k$, $kx + 2y = 12$ is consistent for
- (a) $k = 3$ (b) $k = -3$ (c) $k = 2$ (d) $k = -2$
34. If $a > b > c$ and the system of equations $ax + by + cz = 0$, $bx + cy + az = 0$, $cx + ay + bz = 0$ has a non trivial solution then both the roots of the quadratic equation $at^2 + bt + c = 0$ are
- (a) real (b) of opposite sign (c) positive (d) complex
35. The system of equations $x \sin \theta - \sqrt{3}y = 0$, $x \cos \theta + y = 0$ has infinite many solutions, for θ equals
- (a) $\pi/3$ (b) $2\pi/3$ (c) $5\pi/3$ (d) $4\pi/3$
36. If $\begin{vmatrix} x^2 + x & x+1 & x-2 \\ 2x^2 + 3x - 1 & 3x & 3x-3 \\ x^2 + 2x + 3 & 2x-1 & 2x-1 \end{vmatrix} = Ax + B$. Then
- (a) $A = 12$ (b) $A = 24$ (c) $B = -12$ (d) $B = 12$
37. Let $f_r(x), g_r(x), h_r(x)$; $r = 1, 2, 3$ be polynomials in x such that $f_r(a) = g_r(a) = h_r(a)$. If $\frac{dF(x)}{dx} = F'(x)$ and $F(x) = \begin{vmatrix} f_1(x) & f_2(x) & f_3(x) \\ g_1(x) & g_2(x) & g_3(x) \\ h_1(x) & h_2(x) & h_3(x) \end{vmatrix}$ then
- (a) $F(a) = \text{a constant}$ (b) $F'(a) = \text{a constant}$ (c) $F(a) = 0$ (d) $F'(a) = 0$
38. If $f(x) = \begin{vmatrix} x^n & \sin x & \cos x \\ n! & \sin \frac{n\pi}{2} & \cos \frac{n\pi}{2} \\ k & k^2 & k^3 \end{vmatrix}$ then $\frac{d^n f(x)}{dx^n}$ at $x = 0$
- (a) 0 (b) k (c) k^3 (d) Independent of k

39. If $f(x) = \begin{vmatrix} (1+x)^{a_1 b_1} & (1+x)^{a_1 b_2} \\ (1+x)^{a_2 b_1} & (1+x)^{a_2 b_2} \end{vmatrix}$ then coefficient of x in $f(x)$ is $(a_1 > a_2, b_1 > b_2)$
- (a) 0 (b) a +ve number (c) a -ve number (d) $(a_1 - a_2)(b_1 - b_2)$
40. If $\Delta = \begin{vmatrix} a & a^2 & 0 \\ 1 & 2a+b & (a+b) \\ 0 & 1 & 2a+3b \end{vmatrix}$ is divisible by
- (a) $a+b$ (b) $a+2b$ (c) $2a+3b$ (d) a^2

COMPREHENSION

Passage-1

If $A = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, evaluate three roots $\{\lambda_1, \lambda_2, \lambda_3\}$ of λ from the equation $|A - \lambda I| = 0$ construct a null invertible matrix X such that $AX = \lambda X$ where $\lambda = \text{diagonal } \{\lambda_1, \lambda_2, \lambda_3\}$.

- The three roots of λ are
(a) 1, 0, 2 (b) -1, 1, 3 (c) 2, -2, 3 (d) 5, 1, -3
- Trace of matrix A^n is
(a) $3^n + 1 + 1 + (-1)^n$ (b) $2^n + 1$ (c) $3^n - 2^n + 2$ (d) $3^n + 2^n - 2$
- Matrix a satisfies
(a) $A^2 - (\lambda_1 + \lambda_2)A + \lambda_1 \lambda_2 I = 0$ (b) $A^2 - (\lambda_1 + \lambda_3)A + \lambda_1 \lambda_3 I = 0$
(c) $A^2 - (\lambda_2 + \lambda_3)A + \lambda_2 \lambda_3 I = 0$ (d) none of these

Passage-2

Let $A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{bmatrix}$. If U_1, U_2, U_3 are column matrices

Satisfying $AU_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $AU_2 = \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix}$, $AU_3 = \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}$ and U is 3×3 matrix whose columns are U_1, U_2 and U_3 then

- $|U| =$
(a) 3 (b) $3/2$ (c) -3 (d) 2
- Sum of all elements of $U^{-1} =$
(a) -1 (b) 0 (c) 1 (d) 3
- $[3 \ 2 \ 0] U \begin{bmatrix} 3 \\ 2 \\ 0 \end{bmatrix} =$
(a) 5 (b) $5/2$ (c) 4 (d) $3/2$

Passage-3

Let A and B are two matrices of same order i.e. 3×3 where $A = \begin{bmatrix} 1 & -3 & 2 \\ 2 & K & 5 \\ 4 & 2 & 1 \end{bmatrix}$; $B = \begin{bmatrix} 2 & 1 & 3 \\ 4 & 2 & 4 \\ 3 & 3 & 5 \end{bmatrix}$

On the basis of above information, answer the following questions

- If A is singular matrix then $\text{tr}(A + B)$ is equal to
(a) 5 (b) 3 (c) 4 (d) 6

8. If matrix $2A + 3B$ is singular, then the value of K is
 (a) -3 (b) $-\frac{40}{17}$ (c) $-\frac{42}{13}$ (d) none of these
9. If $K = 2$ then $\text{tr}(AB) + \text{tr}(BA)$ is equal to
 (a) 66 (b) 42 (c) 84 (d) 63
10. If $C = A - B$ and $\text{tr}(C) = 0$ then K is equal to
 (a) 5 (b) -5 (c) 7 (d) -7

Passage-4

If A and B are square matrix of order 3 given by $A = \begin{bmatrix} 1 & 2 & 4 \\ 4 & 3 & 5 \\ 2 & 4 & 6 \end{bmatrix}$; $B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$

On the basis of above informant, answer the following questions

11. $\text{adj}(\text{adj } A)$ is equal to
 (a) 100 A (b) 10 A (c) 1000 A (d) none of these
12. $|\text{adj}(\text{adj } A)|$ is equal to
 (a) 10^2 (b) 100^3 (c) 10^4 (d) none of these
13. $|\text{adj}(AB)|$ is equal to
 (a) 100 (b) 1000 (c) 10^4 (d) none of these

Passage-5

The system of equations $a_1x + b_1y + c_1z = d_1$, $a_2x + b_2y + c_2z = d_2$, $a_3x + b_3y + c_3z = d_3$ can be written as $AX = B$

where $A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ and $B = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}$ the system is

- (i) consistent with unique solution iff $|A| \neq 0$
 (ii) either inconsistent or consistent with infinite solutions if $|A| = 0$ and $(\text{adj } A) B = 0$
 (iii) Inconsistent iff $|A| = 0$ and $(\text{adj } A) B \neq 0$
14. The system of equations $x + y + z = 3$, $2x + y + 2z = 5$, $x - y + 3z = 3$ has
 (a) only solution $x = 1, y = 1, z = 1$ (b) Infinite solutions
 (c) no solution (d) none of these
15. The system of equations $x + y + 2z = 3$, $3x + 3y + 6z = 12$, $4x + 4y + 8z = 16$ has
 (a) only solution $x = 2, y = 1, z = 1$ (b) Infinite solution
 (c) no solution (d) none of these
16. The system of equations $x + y + z = 3$, $2x + 2y + 2z = 7$, $x - y + 3z = 3$ has
 (a) only solution $x = 1, y = 1, z = 1$ (b) Infinite solution
 (c) no solution (d) none of these

Passage-6

A determinant is called cyclic if it follows the arrangement symmetrically with a, b, c for example $\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix}$ is

a cyclic determinant

Now, if we increase the degree of any row in this determinant symmetrically its value will be multiplied by expression which is also cyclic and increases the degree of the value of determinant,

17. The value of $\begin{vmatrix} 1 & 1 & 1 \\ a^2 & b^2 & c^2 \\ bc & ca & ab \end{vmatrix}$ is equal to
 (a) $(a-b)(b-c)(c-a)$ (b) $(a-b)(b-c)(c-a)(a+b+c)$
 (c) $(a-b)(b-c)(c-a)(ab+bc+ca)$ (d) $(a-b)(b-c)(c-a)abc$

18. The value of $\begin{vmatrix} a & b & c \\ a^2 & b^2 & c^2 \\ bc & ca & ab \end{vmatrix}$ is equal to
- (a) $(a-b)(b-c)(c-a)(a+b+c)^2$ (b) $(a-b)(b-c)(c-a)(a+b+c)$
 (c) $(a-b)(b-c)(c-a)(ab+bc+ca)$ (d) $(a-b)(b-c)(c-a)(a^2+b^2+c^2+ab+bc+ca)$
19. The value of $\begin{vmatrix} a & b & c \\ a^2 & b^2 & c^2 \\ a^3 & b^3 & c^3 \end{vmatrix}$ is equal to
- (a) $(a-b)(b-c)(c-a)(a^3+b^3+c^3-abc)$ (b) $(a-b)(b-c)(c-a)(a+b+c)$
 (c) $(a-b)(b-c)(c-a)(ab+bc+ca)$ (d) $(a-b)(b-c)(c-a)abc$

MATRIX-MATCH

Matrix-1

1. Match the following column :

	Column I		Column II
(a)	If $f(x) = \begin{vmatrix} x & \sin x & \cos x \\ x^2 & -\tan x & -x^3 \\ 2x & \sin 2x & 5x \end{vmatrix}$, then $\lim_{x \rightarrow 0} \frac{f'(x)}{x} =$	(p)	4
(b)	If $x \neq 0, y \neq 0, z \neq 0$ and $\begin{bmatrix} 1+x & 1+y & 1+z \\ 1 & 1+2y & 1+z \\ 1 & 1 & 1+3z \end{bmatrix} = 0$, then $-\left(\frac{1}{x} + \frac{1}{y} + \frac{1}{z}\right) =$	(q)	1
(c)	Let $f(x) = \begin{vmatrix} x & 1 & 1 \\ \sin \pi x & 2x^2 & 1 \\ x^3 & 3x^4 & 1 \end{vmatrix}$ If $f(x)$ be an odd function and its odd Value is equal to $g(x)$, then $f(1)g(1)$ is	(r)	3
(d)	If the system of equations $-ax + y + z = 0, x - by + z = 0$ and $x + y - cz = 0$ ($a, b, c \neq -1$) has a non-zero solution, then $\frac{1}{1+a} + \frac{1}{1+b} + \frac{1}{1+c} =$	(s)	-4

Matrix-2

2. Match the following column :

	Column I		Column II
(a)	The inverse of a skew-symmetric matrix (if it exists) is	(p)	neither symmetric nor orthogonal nor skew-symmetric
(b)	If A is any orthogonal matrix, then $ A =$	(q)	symmetric
(c)	If A is an invertible matrix and B is an orthogonal matrix of the same order as that of A, then $C = A^{-1}BA$ is	(r)	skew-symmetric
(d)	Inverse of a symmetric matrix (if it exists) is	(s)	± 1

Matrix-3

3. Let
$$\begin{vmatrix} 1+x & x & x^2 \\ x & 1+x & x^2 \\ x^2 & x & 1+x \end{vmatrix} = ax^5 + bx^4 + cx^3 + dx^2 + ex + f$$

Observe the following columns :

	Column I		Column II
(a)	The value of f is equal to	(p)	A is Idempotent
(b)	The value of e is equal to	(q)	1
(c)	The value of $a+c$ is equal to	(r)	-1
(d)	The value of $b+d$ is equal to	(s)	3

Matrix-4

4. Observe the following columns :

	Column I		Column II
(a)	If matrix $A = \begin{bmatrix} 1 & 0 & -1 \\ 3 & 4 & 5 \\ 0 & 6 & 7 \end{bmatrix}$ and its inverse $A^{-1} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$ then the value of a_{23} is	(p)	$-\frac{2}{5}$
(b)	If $A = \begin{bmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{bmatrix}$ and $A (\text{Adj. } A) = k \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ then the value of k is	(q)	1
(c)	If $A = \begin{bmatrix} 1 & \log_b a \\ \log_a b & 1 \end{bmatrix}$ then $ A $ is equal to	(r)	0
(d)	If $A = \begin{bmatrix} a & b \\ 0 & a \end{bmatrix}$ is n th root of $4I_2$, then the value of a^n is	(s)	4

Matrix-5

5. Observe the following columns :

	Column I		Column II
(a)	If ω is a cube root of unity, then a root of the equation $\begin{vmatrix} x+1 & \omega & \omega^2 \\ \omega & x+\omega^2 & 1 \\ \omega^2 & 1 & x+\omega \end{vmatrix} = 0$ is	(p)	0
(b)	If $a=i, b=\omega$ and $c=\omega^2$, then the value of determinant $\begin{vmatrix} a & a+b & a+b+c \\ 3a & 4a+3b & 5a+4b+3c \\ 6a & 9a+6b & 11a+9b+6c \end{vmatrix}$ is	(q)	$-3\sqrt{3}i$
(c)	The value of the determinant $\begin{vmatrix} 1 & 1 & 1 \\ 1 & \omega^2 & \omega \\ 1 & \omega & \omega^2 \end{vmatrix}$ is equal to	(r)	i
(d)	Let a, b, c are real and different such that $\Delta = \begin{vmatrix} \sum a^2 & \sum ab & \sum ab \\ \sum ab & \sum a^2 & \sum ab \\ \sum ab & \sum ab & \sum a^2 \end{vmatrix}$	(s)	$3\sqrt{3}i$

Matrix-6

6. Observe the following columns :

	Column I		Column II
(a)	A is square matrix such that $A^2 = A$	(p)	A is Nipotent matrix
(b)	A is a square matrix such that $A^m = 0$	(q)	A is Involutary matrix
(c)	A is a square matrix such that $A^2 = I$	(r)	A is Symmetric matrix
(d)	A is a square matrix such that $A^T = A$	(s)	A is Idempotent matrix

NUMERICAL GRID

- Find the rank of the matrix $\begin{bmatrix} 1^2 & 2^2 & 3^2 & 4^2 \\ 2^2 & 3^2 & 4^2 & 5^2 \\ 3^2 & 4^2 & 5^2 & 6^2 \\ 4^2 & 5^2 & 6^2 & 7^2 \end{bmatrix}$
- Evaluate : $A = \begin{vmatrix} 1 & \log_x y & \log_x z \\ \log_y x & 1 & \log_y z \\ \log_z x & \log_z y & 1 \end{vmatrix}$ where x, y, z being +ve.
- If T_p be the p th term of a G.P. of +ve term, then evaluate $\begin{vmatrix} \log T_{p+1} & \log T_{p+3} & \log T_{p+5} \\ \log T_{p+3} & \log T_{p+5} & \log T_{p+7} \\ \log T_{p+5} & \log T_{p+7} & \log T_{p+9} \end{vmatrix}$
- Find the value of n for which the determinant $\Delta = \begin{vmatrix} {}^8C_3 & {}^9C_5 & {}^{10}C_7 \\ {}^8C_4 & {}^9C_6 & {}^{10}C_8 \\ {}^8C_n & {}^9C_{n+2} & {}^{10}C_{n+4} \end{vmatrix} = 0$
- If ω be imaginary cube root of unity then evaluate $\begin{vmatrix} 2+3\omega^{100}+\omega^{203} & 2\omega^2 & 1 \\ 1 & 1+\omega^{100}+3\omega^{203} & 2\omega \\ 2\omega & 2\omega^2 & 3+\omega^{100}+\omega^{203} \end{vmatrix}$
- If $x \neq 0, y \neq 0, z \neq 0$ and $\begin{vmatrix} 1+x & 1 & 1 \\ 1+y & 1+2y & 1 \\ 1+z & 1+z & 1+3z \end{vmatrix} = 0$, then $-(x^{-1} + y^{-1} + z^{-1})$ is equal to
- If $\Delta = \begin{vmatrix} f(x) & f\left(\frac{1}{x}\right) + f(x) \\ 1 & f\left(\frac{1}{x}\right) \end{vmatrix} = 0$ where it is given, $f(x) = a + bx^n$ and $f(2) = 17$, then $f(5)$ is equal to :
- If $f(x) = \begin{vmatrix} x & \sin x & \cos x \\ x^2 & -\tan x & -x^3 \\ 2x & \sin 2x & 5x \end{vmatrix}$ then $\lim_{x \rightarrow 0} \frac{f'(x)}{x}$ is equal to
- If $3A = \begin{vmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ a & 2 & b \end{vmatrix}$ and A is an orthogonal matrix then the product of a and b is equal to
- The set of natural numbers N is partitioned into arrays of rows and columns in the form of matrices as $M_1 = (1)$, $M_2 = \begin{pmatrix} 2 & 3 \\ 4 & 5 \end{pmatrix}$, $M_3 = \begin{pmatrix} 6 & 7 & 8 \\ 9 & 10 & 11 \\ 12 & 13 & 14 \end{pmatrix}$,, $M_n = ()$ and so on.
Find the sum of the elements of the diagonal in M_6 .

11. If $A = \begin{bmatrix} 0 & c & -b \\ -c & 0 & a \\ b & -a & 0 \end{bmatrix}$ and $B = \begin{bmatrix} a^2 & ab & ac \\ ba & b^2 & bc \\ ca & cb & c^2 \end{bmatrix}$ then AB is equal to
12. If $D_k = \begin{vmatrix} 1 & n & n \\ 2k & n^2 + n + 2 & n^2 + n \\ 2k-1 & n^2 & n^2 + n + 2 \end{vmatrix}$ and $\sum_{k=1}^n D_k = 48$, then n is equal to
13. Number of values of θ in $(-3\pi, \pi)$ for which the system $(\cos \theta)x - y = 0$, $x + 2y = 0$ has infinite number of solutions is
14. If a, b, c are non zero real numbers and the system of equations $a(x + y + z) = x$; $b(x + y + z) = y$; $c(x + y + z) = z$ has a non-trivial solution, then $a + b + c$ equals
15. The number of values of k for which the system of equations $(k+1)x + 8y = 4k$ and $kx + (k+3)y = 3k - 1$ has infinitely many solutions is
16. If $A = \begin{bmatrix} 1 & -1 \\ 2 & -1 \end{bmatrix}$, $B = \begin{bmatrix} x & 1 \\ y & -1 \end{bmatrix}$ and $(A+B)^2 = A^2 + B^2$, then $x + y =$
17. If $A = \begin{bmatrix} 2^2 & 3^2 & 4^2 \\ 3^2 & 4^2 & 5^2 \\ 4^2 & 5^2 & 6^2 \end{bmatrix}$ then find $|A|$
18. If x, y, z are in A.P., then the value of the determinant $\begin{vmatrix} a+2 & a+3 & a+2x \\ a+3 & a+4 & a+2y \\ a+4 & a+5 & a+2z \end{vmatrix}$ is
19. If x_1, x_2 and y_1, y_2 are the roots of the equations $3x^2 - 18x + 9 = 0$ and $y^2 - 4y + 2 = 0$, then the value of determinant $\begin{vmatrix} x_1x_2 & y_1y_2 & 1 \\ x_1 + x_2 & y_1 + y_2 & 2 \\ \sin(\pi x_1 x_2) & \cos\left(\frac{\pi}{2} x_1 x_2\right) & 1 \end{vmatrix}$ is
20. If A, B, C are the angles of a triangle then the value of determinant $\begin{vmatrix} \sin 2A & \sin C & \sin B \\ \sin C & \sin 2B & \sin A \\ \sin B & \sin A & \sin 2C \end{vmatrix}$ is
21. If $f(x) = \begin{vmatrix} \cos(x+\alpha) & \cos(x+\beta) & \cos(x+\gamma) \\ \sin(x+\alpha) & \sin(x+\beta) & \sin(x+\gamma) \\ \sin(\beta-\gamma) & \sin(\gamma-\alpha) & \sin(\alpha-\beta) \end{vmatrix}$ and $f(2) = 5$, then $\sum_{r=1}^{20} f(r)$ equals
22. If a, b, c are in A.P. and $f(x) = \begin{vmatrix} x+a & x^2+1 & 1 \\ x+b & 2x^2-1 & 1 \\ x+c & 3x^2-1 & 1 \end{vmatrix}$, then $f'(x)$ is
23. If $f(x) = \begin{vmatrix} \cos(x+x^2) & \sin(x+x^2) & -\cos(x+x^2) \\ \sin(x-x^2) & \cos(x-x^2) & \sin(x-x^2) \\ \sin 2x & 0 & \sin 2x^2 \end{vmatrix}$, then $f'(0)$

ANSWER KEY

LEVEL-1

DAY-1

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
c	a	a	d	a	c	d	c	c	a	b	d	b	d	d	b	c	c	c	b	c	b	a	a	c	d	c	a	b	c

DAY-2

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
b	a	c	c	c	c	c	a	a	c	c	d	a	d	d	a	b	a	d	d
21	22	23	24	25	26	27	28	29	30	31	32	33	34	35					
b	d	a	c	b	d	b	a	b	a	d	a	b	c	d					

DAY-3

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
c	d	d	d	d	c	a	b	b	b	b	b	d	a	a	b	a	a	b	d

DAY-4

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
a	c	a	a	a	d	b	c	d	c	b	d	d	d	b	c	c	a	d	d
21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
a	d	b	c	c	a	d	c	b	c	d	d	b	b	d	d	c	d	c	b

DAY-5

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
b	a	b	c	c	a	b	d	a	a	d	b	b	c	c	b	c	b	d	b
21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
d	a	a	d	b	d	c	a	d	a	b	d	a	c	c	b	ab	b	c	c

DAY-6

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
a	b	c	a	a	d	c	b	d	a	c	d	c	b	a	d	d	a	a	b	a	a	c	c	B

DAY-7

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
a	a	b	b	d	a	b	d	a	b	b	b	a	c	b	c	a	b	b	d	d	d	a	b	d

DAY-8

1	2	3	4	5	6	7	8	9	10
c	b	b	d	b	a	d	b	a	b

DAY-9

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
a	a	d	b	a	a	a	a	a	a	b	c	c	c	b	d	b	a	b	c

LEVEL-2 : (SINGLE CORRECT OPTION)

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
c	c	c	c	a	d	c	d	a	a	d	b	d	a	a	d	a	b	a	a	b	b	a	c	a
26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50
a	a	c	c	b	d	a	d	c	a	a	a	c	a	a	c	d	b	c	d	b	d	c	d	c
51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
c	a	a	c	a	d	b	b	c	c	a	b	a	c	c	d	d	d	b	b	c	b	c	a	c

LEVEL-3 (MULTIPLE CORRECT OPTIONS)

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
ab	ac	cd	ac	ab	ab	abc	ac	abc	ab	abcd	abcd	abd	acd	abcd	acd	abcd	bc	ab	abcd
21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
b	ab	bc	ac	c	acd	abc	abcd	ad	ac	acd	ac	ab	ab	bc	bc	abc	ad	bd	ab

PASSAGE :

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
b	d	d	a	b	a	b	b	c	c	b	d	d	a	c	c	b	c	d

MATRIX-MATCH

1	2	3	4	5	6
p, r, s, q	r, s, p, q	q, s, r, q	p, q, r, s	p, r, q, p	s, p, q, r

NUMERICAL GRID

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
3	0	0	4	0	3	626	4	2	441	0	4	4	1	1	5	-8	0	0	0	10	0	2